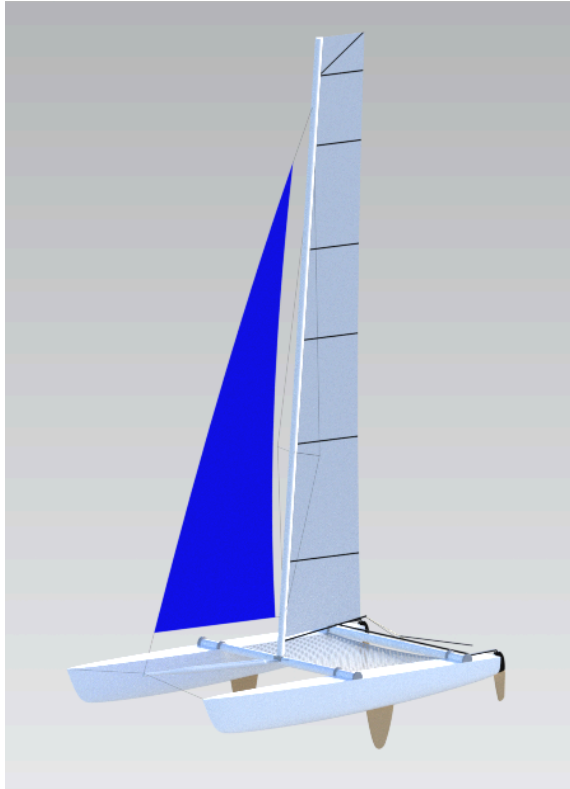


RACING CATAMARAN STRESS AND DYNAMIC ANALYSIS



A ME4042 final project

by

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CHAPTER 1. INTRODUCTION

A catamaran is a twin-hulled vessel designed for stability, efficiency, and speed. This project focuses on developing a 3D CAD model of a catamaran, followed by finite-element analyses (FEA) and computational fluid dynamics (CFD) to evaluate its structural and aerodynamic performance under representative marine conditions.

1.1 Motivation and objectives

Unlike monohulls, catamarans distribute loads across two slender hulls connected by a deck and truss structure, resulting in reduced hydrodynamic drag and improved stability in rough waters. This special geometry of a catamaran leads to an increased engineering complexity which in turn allows for unique structural and aerodynamic analysis. For instance, when “flying a hull” (one hull out of the water), drag and stability change dramatically; crossbeams and stays see elevated stresses from crew counter-moment(trapeze) and sail forces.

This project will create a Siemens NX model and perform an integrated analysis: FEA for structural and fatigue performance under realistic sailing loads, CFD for drag at maximum speed and heel angles, and a material/geometry optimization to balance strength, stiffness, mass, and hydrodynamic performance.

1.2 Parametric model overview

The vessel used in this study was modeled after the International Tornado, which is a high-performance, two-person catamaran originally designed for The Olympics. This decision

was due to the Tornado's popularity which provides many online resources to assist with modeling and results validation, for an arguably niche topic.

The Catamaran is composed of 4 main components: twin-hulls, crossbeams, mast, and sails. Minor components, such as the daggerboards, which prevent lateral drift, rudders, which steer the vessel, and rigging system to control the sails and mast, were also modeled and discussed in detail here – see Chapter 2.

1.3 Proposed analysis and failure criteria

For the finite-element analyses (FEA) and computational fluid dynamics (CFD), we plan to conduct several studies to evaluate and optimize the structural, hydrodynamic, and aerodynamic performance of the catamaran under realistic operating conditions.

First, we will simulate extreme environmental loading, such as high-speed winds during severe weather, to calculate the stresses and deformations on the sails and mast, ensuring that the design can withstand peak aerodynamic forces. Second, we will analyze the vessel's structural behavior while flying a hull. For this, the catamaran will be approximated as cantilevered on a single hull to study the structural response of the connecting trusses and joints, computing stresses, strains, and potential failure modes under asymmetric loading.

Finally, we will consider heel angle versus drag. The heel angle is defined as the angle which the catamaran's base makes with the surface of the water. This will involve performing parametric simulations to determine the hull and deck geometries that result in the lowest drag at various heel angles, including identifying the optimal angle for minimal resistance – unique to

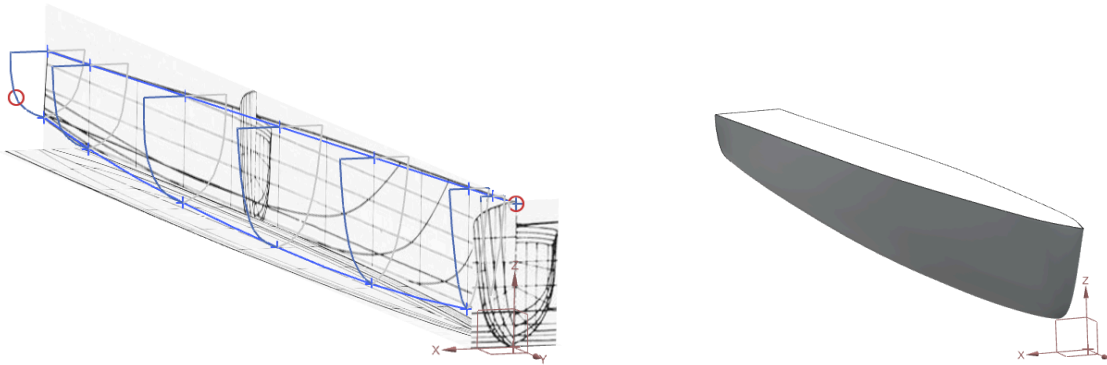
each catamaran. Since this is highly dependent on the vessel's design and material composition, our results will likely vary from those reported in literature.

These combined analyses will provide a comprehensive understanding of both the hydrodynamic efficiency and structural integrity of the catamaran, guiding subsequent geometry and material optimization.

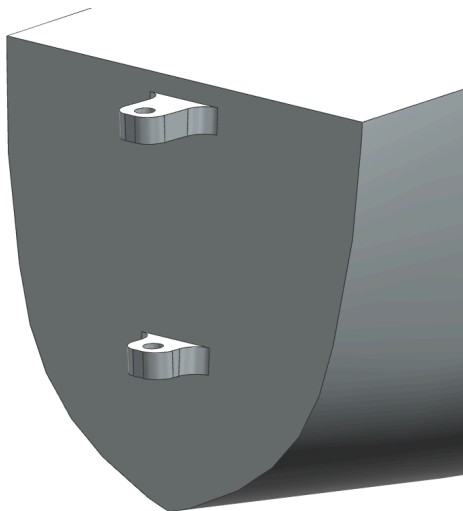
CHAPTER 2. PARAMETRIC MODEL AND MATERIALS SELECTION

The following chapter details the development of the catamaran's parametric CAD model and the rationale behind the materials chosen for each major component. A fully parametric approach was adopted in Siemens NX to allow rapid iteration of hull geometry, crossbeam dimensions, mast profile, and sail shape, enabling efficient optimization during FEA and CFD studies. Material selection was guided by mechanical performance, weight, durability in marine environments, manufacturability, and alignment with high-performance racing standards. By integrating geometry and material decisions early in the design process, we ensure that subsequent structural and aerodynamic analyses accurately reflect realistic loading behavior and enable a performance-driven optimization of the vessel.

2.1 Hulls

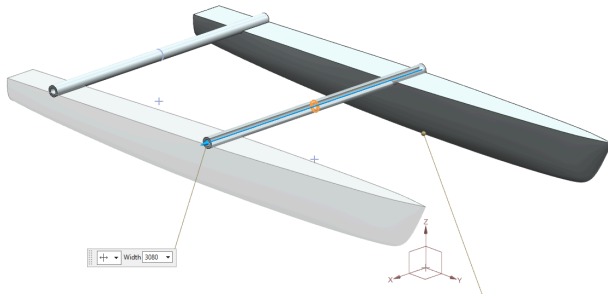


Guide curves of the hull's shape and cross sections along the hull were sketched using a raster image. Through curves was used by selecting the cross sections and guide curves to create the hulls.

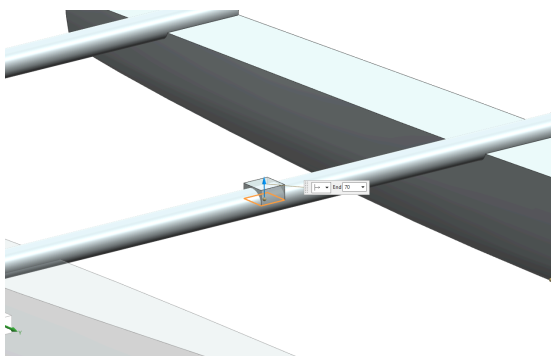


These are extruded out on the back of the hull so the rudders can be attached.

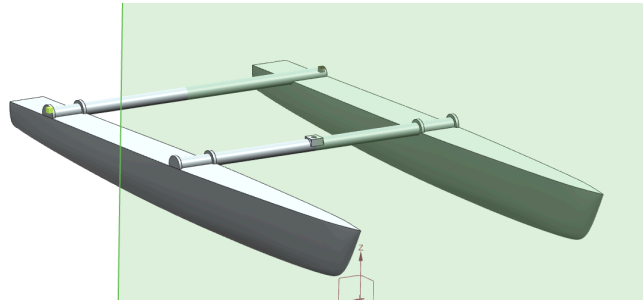
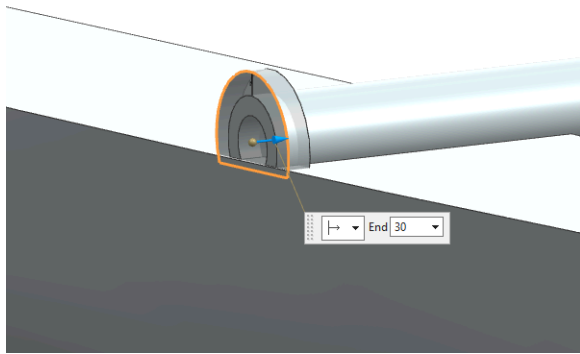
2.2 Crossbeams and deck



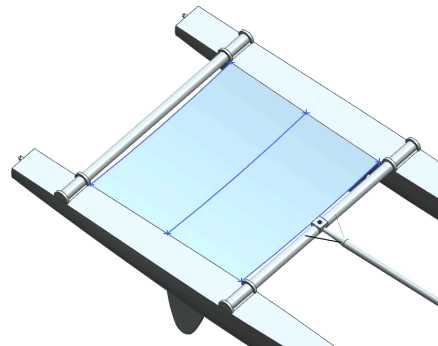
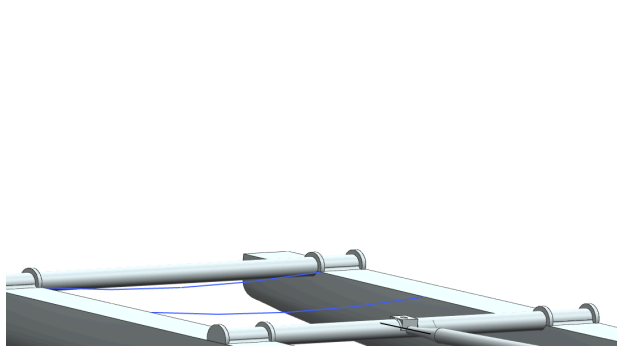
The beam's cross section was sketched on the midplane between the two hulls, and extruded and united to the hulls.



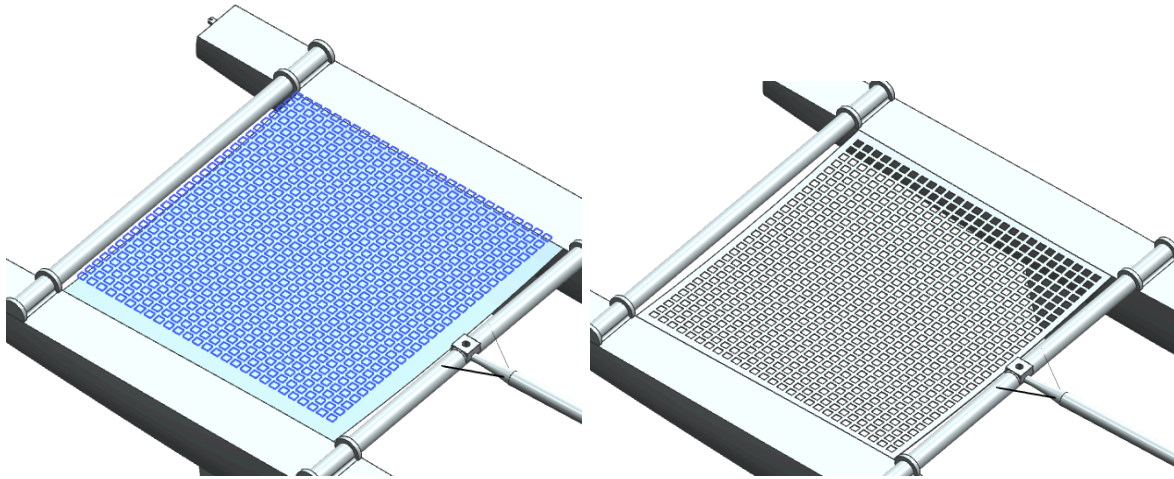
A square is extruded through the front cross beam, and a circular hole is cut through it so the mast can attach to the crossbeams.



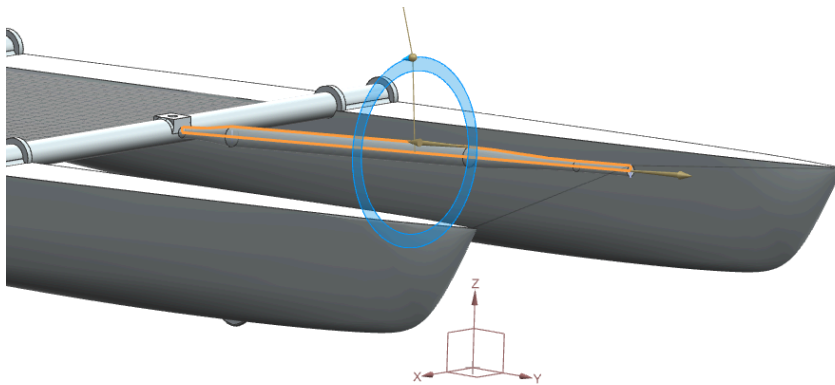
A profile of the bracket that holds the beam in place is sketched and extruded at the edge of the beam. It is mirrored across the midplane of each individual hull, and the midplane between the hulls.



Cross curves are sketched between the hulls for the outline of the deck. Through curve mesh is used to create the surface of the deck, and thickened is used to make it a solid body.

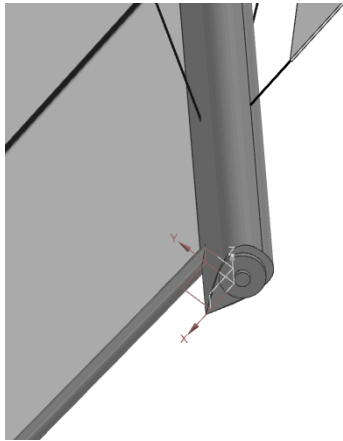


A net pattern is sketched above the surface, and cut through to make the deck.



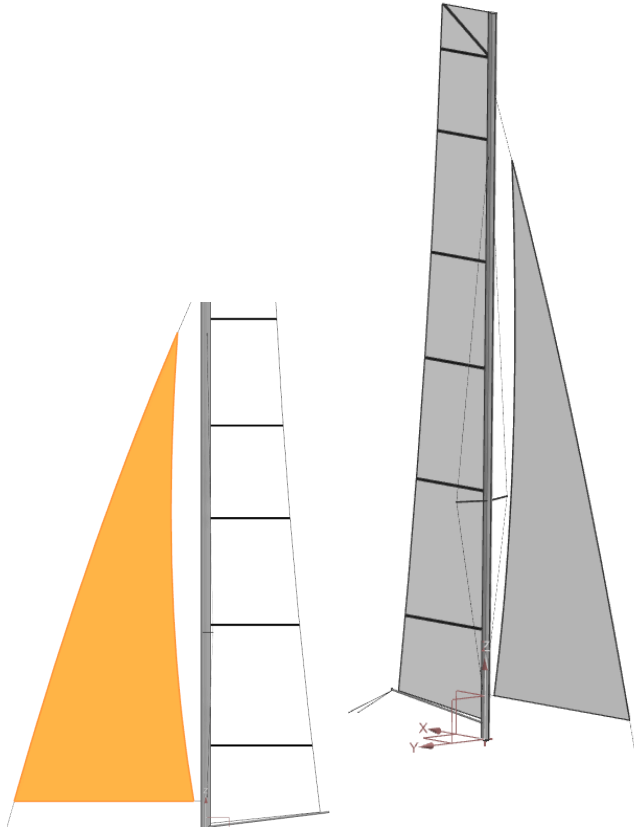
The front stay was modelled by revolving a sketch that extends out of the front crossbeam.

2.3 Mast



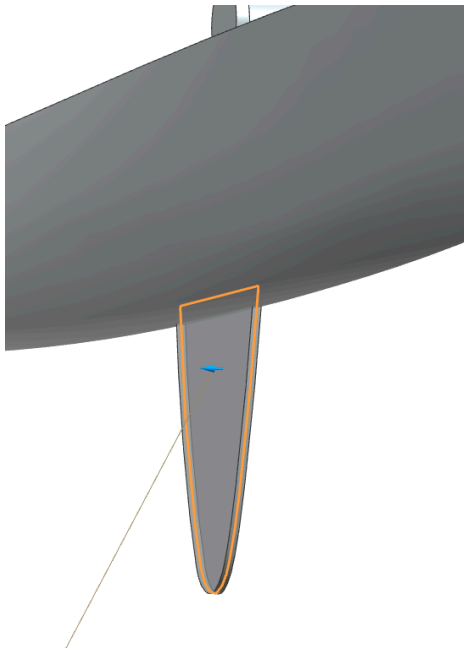
A circle is extruded out the bottom of the mast so it can be assembled with the crossbeams. A teardrop cross section is extruded upwards to create the mast.

2.4 Sails



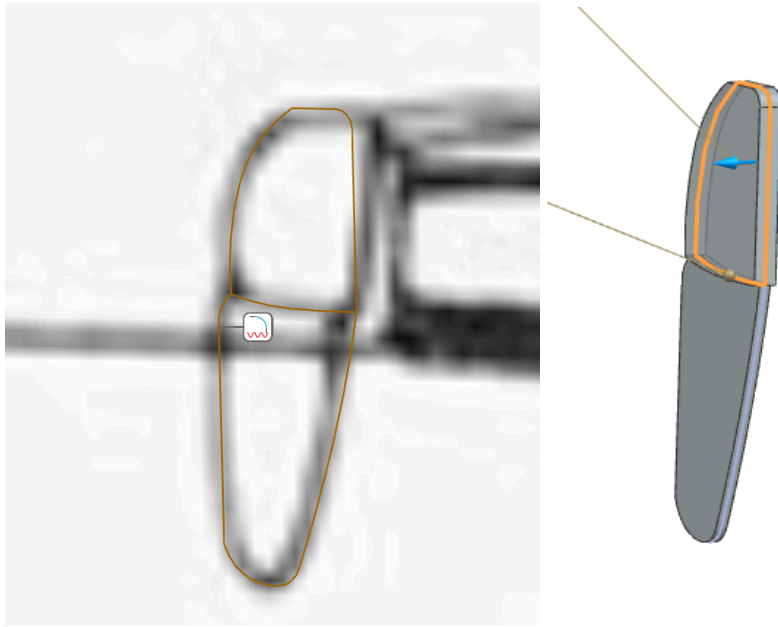
The ribs of the sail were extruded out from the mast. The sails were created using an extrude of their sketched profile.

2.5 Daggerboard

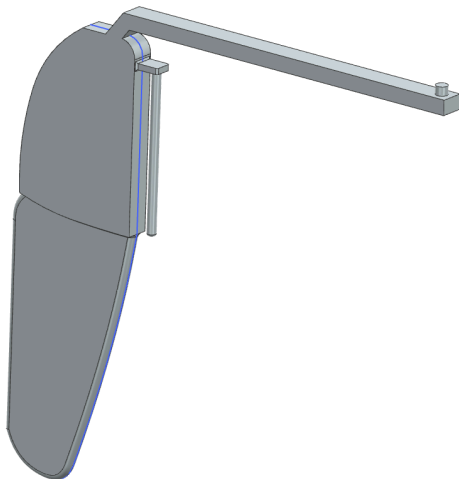


The side profile of the daggerboard is sketched using the raster image and extruded to make the daggerboards.

2.6 Rudders

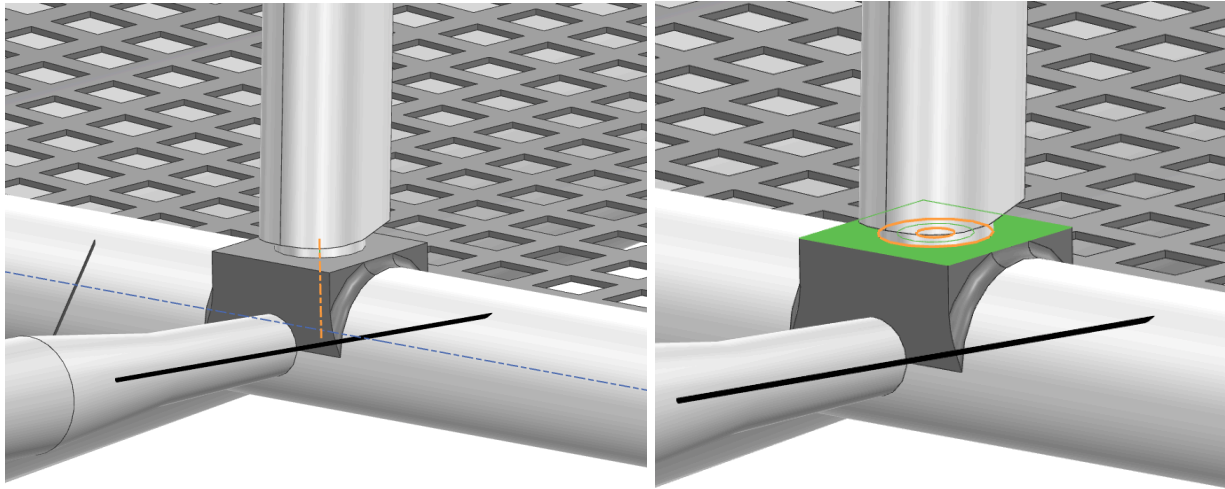


The side profile of the rudders is sketched using the raster image, and extruded to create the model.

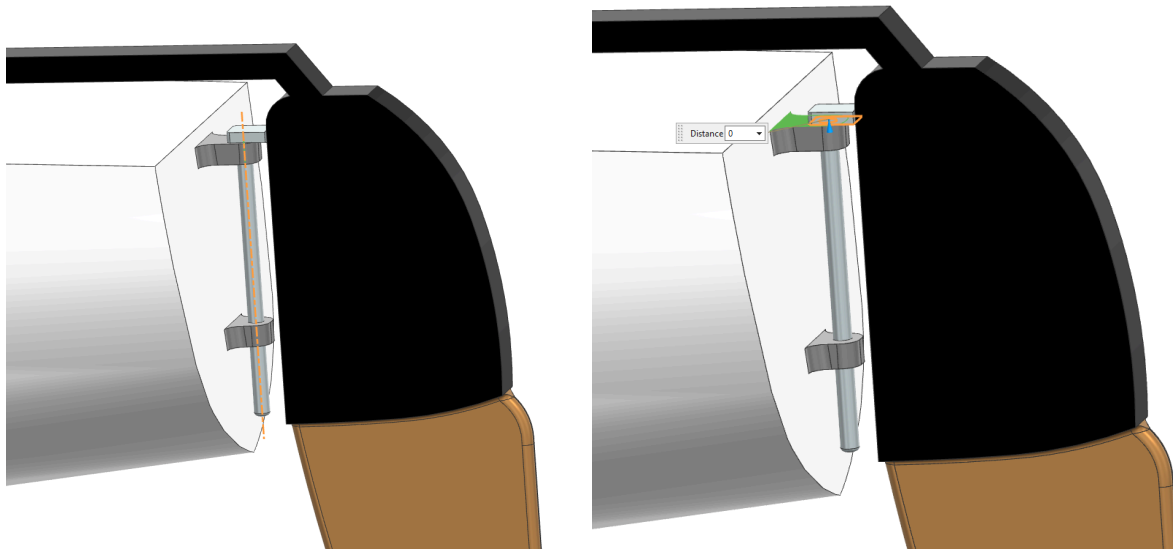


A rod is extruded down and in front of the rudder so it can be attached to the hull, and a beam is extended above and in front so the rudders can be connected by a crossbar.

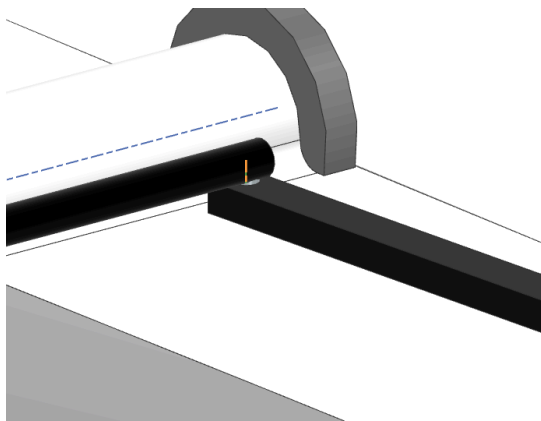
2.7 Assembly



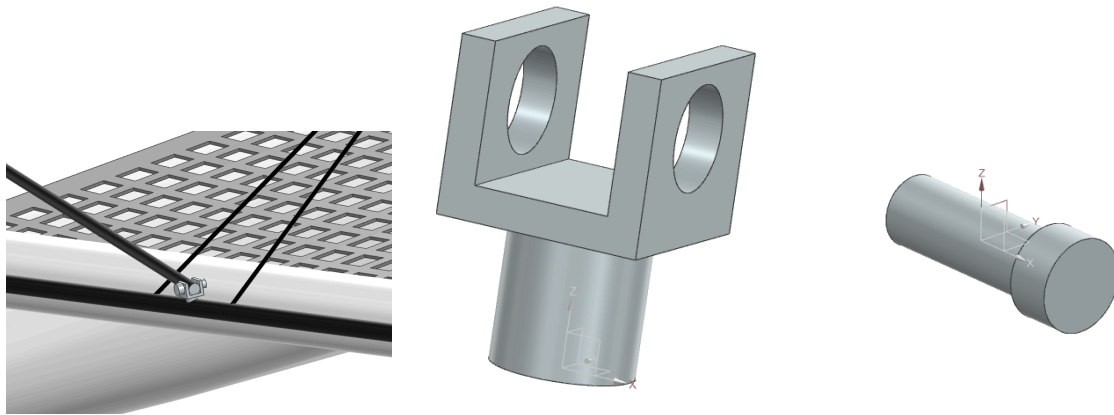
The axes of the cross beam cut out and the mast are aligned using the aligned constraint. The bottom surface of the mast and the top surface of the crossbeam are mated using the touch constraint.



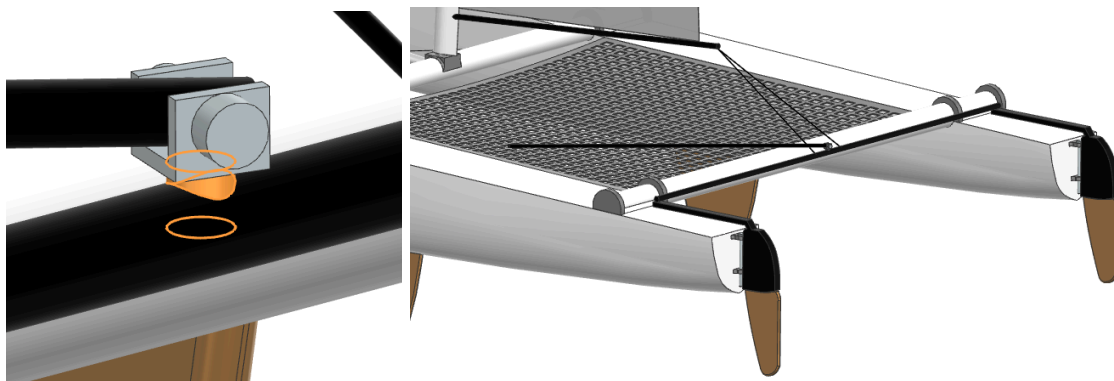
The centerline of the rudder rod and hole in the extrusion in the back of the hull are aligned using the aligned constraint. The top surface of the extrusion out the back of the rudder and top surface of the rudder rod are mated using a distance = 0 constraint.

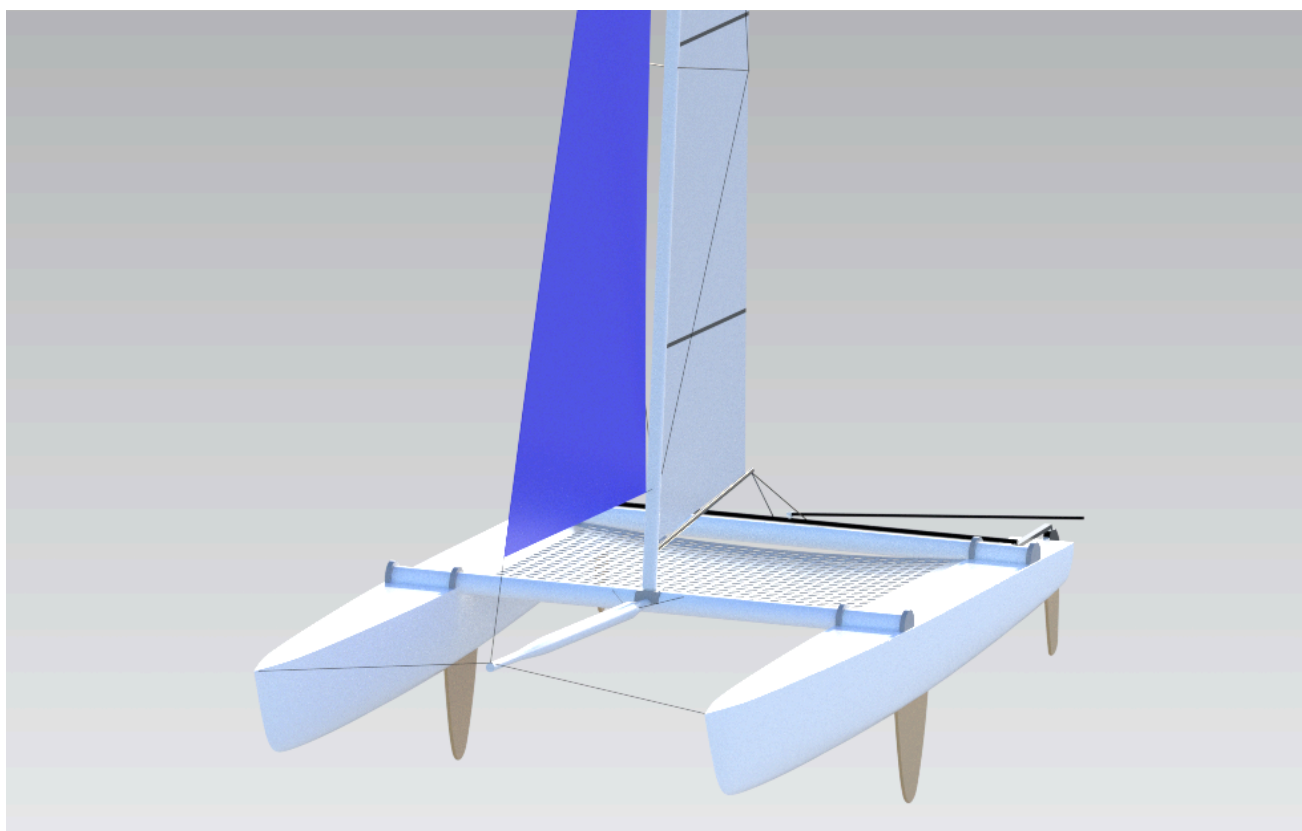
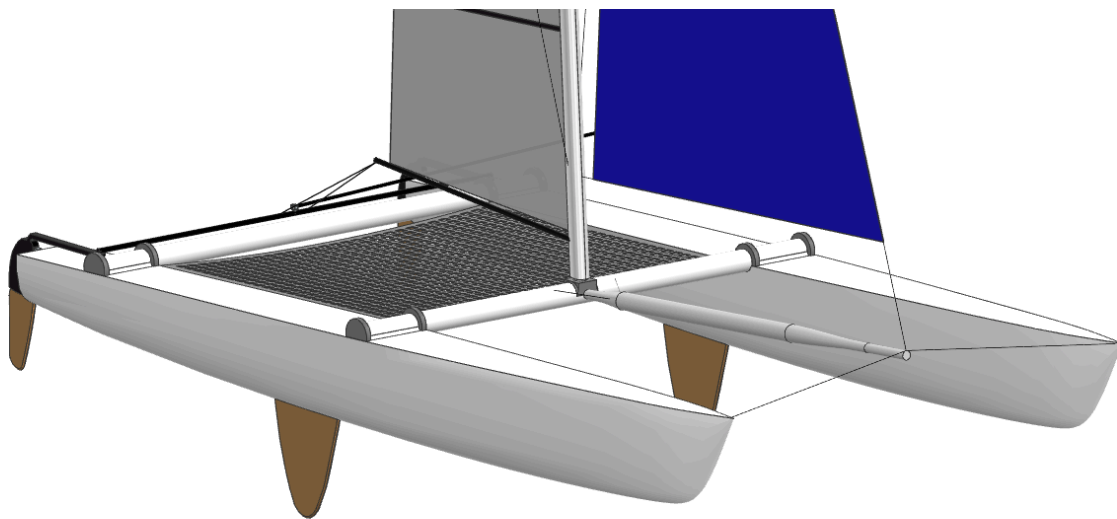


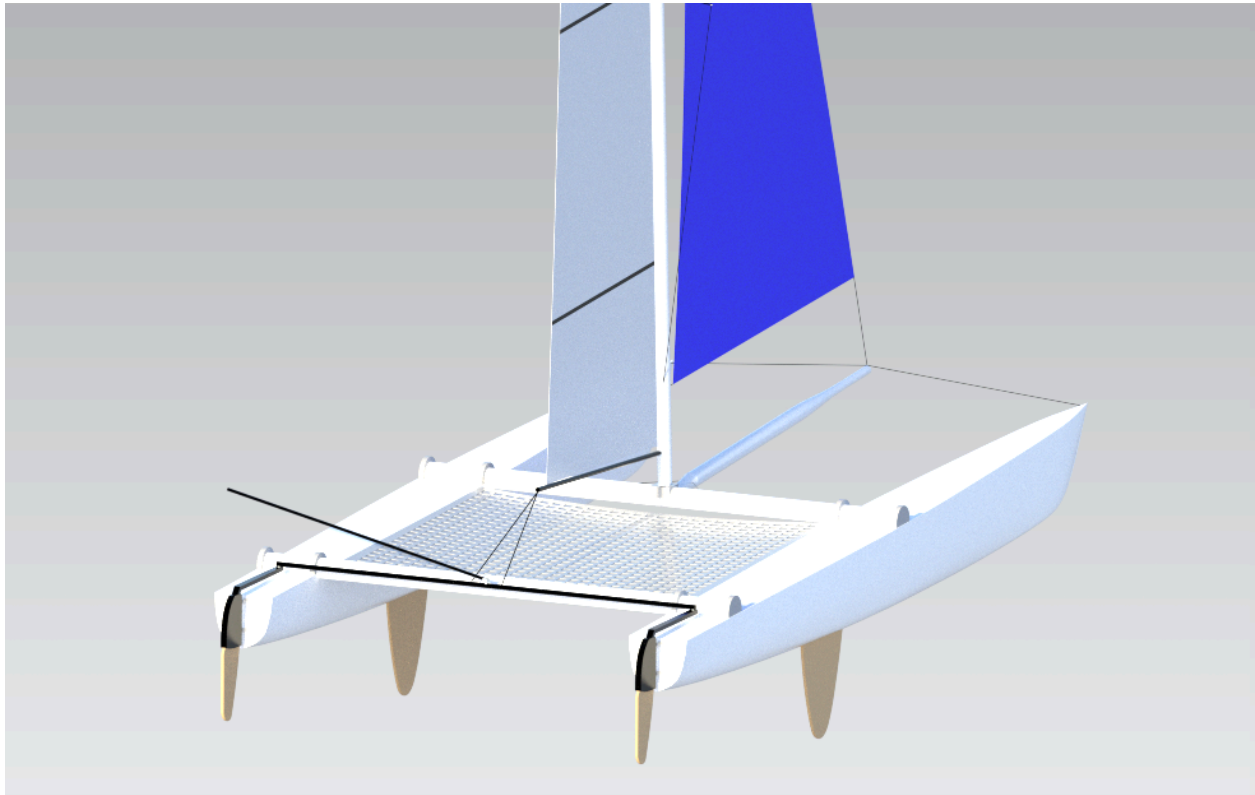
The crossbar is mated to both the rudders using an aligned constraint to align their centerlines.



The steering handle is connected to the crossbar via a bracket and pin.







2.8 Materials Selection

The hulls demand a material system capable of longitudinal bending, transverse shear, torsion from asymmetric loading, and high local pressures around foils and rudders, all the while keeping structural mass extremely low. To meet these multiaxial stress and weight requirements, the hulls are specified as epoxy/HS carbon fiber, resin infused 2x2 twill weave. A quasi-isotropic laminate ($0^\circ/+45^\circ/-45^\circ/90^\circ$) delivers balanced stiffness and strength in all principal directions without compromising on weight. Epoxy provides low water absorption, excellent fatigue resistance, and strong fiber-matrix adhesion. The cross-beams, daggerboards, and rudders also experience multi-axial loads and therefore are assigned the same material.

The mast experiences a more uniaxial loading environment, dominated by longitudinal bending along its length, compressive loads from the standing rigging, and torsional inputs from the mainsail leech. Because the loading is primarily aligned with the mast axis, the same material but using a biaxial layup with strong emphasis on 0°/90° carbon fibers was selected. This configuration maximizes bending stiffness while minimizing flex, allowing the mast to maintain the designed sail shape and reduce aerodynamic drag from mast deformation. Having only changed the composite's layup, the remaining non-mechanical properties, such as corrosion resistance to salt water and UV light, remain satisfactory.

The sailcloth is not merely an aerodynamic surface and must also function as a load-bearing aerodynamic membrane whose performance depends directly on its ability to resist stretch. Even the slightest elastic deformations shift draft position, distort camber, and undermine upwind efficiency. To preserve the designed airfoil geometry in all conditions, the sails are specified in Dyneema-1000, an ultra-high modulus polyethylene fiber with exceptional specific strength, near-zero creep, and outstanding UV and moisture resistance. Compared with other candidate materials, such as Vectran and carbon laminates, Dyneema 1000 maintains both stiffness and durability under repeated flexing, making it ideal for a high-performance multihull where long-term shape retention and fatigue resistance are paramount.

The standing rigging must also maintain constant geometry because any elongation in shrouds or forestay alters mast rake, increases forestay sag, and compromises both balance and aerodynamic efficiency. For this reason, the cables are also specified in Dyneema 1000. Together, the Dyneema sailcloth and rigging create a zero-stretch aerodynamic and structural

system that preserves sail shape, mast alignment, and load-path consistency throughout the vessel.

Component	Material	Properties
Hulls, Crossbeams, Daggerboards, Rudders	Epoxy/HS carbon fibre, QI lay-up [0°/+45°/-45°/90°]	Density = 1.52e3 - 1.54e3 kg/m ³ E = 41 - 46 GPa Yield Strength = 383 - 552 MPa ν = 0.33 - 0.34
Mast	Epoxy/HS carbon fibre, Bi-axial lay-up [0°/90°]	Density = 1.49e3 - 1.54e3 kg/m ³ E = 58 - 64 GPa Yield Strength = 533 - 774 MPa ν = 0.058 - 0.1
Sail	Dyneema 1000 (PE-UHMW)	Density = 965 - 975 kg/m ³ E = 120 -170 GPa Yield Strength = 2.9e3 - 3e3 MPa ν = 0.41- 0.43

Table: Summary of Materials for each component and respective material properties. All values taken from GrantaEduPack, database Level 3.

CHAPTER 3. STRESS ANALYSIS DURING A FRESH GALE

A fresh gale is a severe wind condition defined by sustained winds of approximately 34–40 knots and steep, breaking seas. For a high-performance racing catamaran, a fresh gale represents a critical test of structural integrity: the apparent wind angle tightens dramatically, sail forces increase nonlinearly with wind speed, and dynamic loads propagate through the mast, crossbeams, and hulls. These conditions are intentionally chosen as the design stress case

because they combine the worst of both aerodynamic and hydrodynamic environments while still representing a realistic scenario for competitive multihull sailing.

As the catamaran accelerates, the apparent wind strengthens and shifts forward, creating large bending moments in the mast and increased compression and shear in the stays and forestay. The mainsail and headsail generate high, fluctuating pressure differentials, translating into torsion along the yard and substantial in-plane stresses across the sail membrane. Meanwhile, the hulls encounter significant slamming loads, particularly on the leeward side (i.e., side opposite to the wind direction), while the windward hull may partially lift out of the water, creating highly asymmetric stresses in the crossbeams and deck attachment points. Rudders, daggerboards, and their supporting structures also endure elevated lateral forces as they maintain course against heavy gusts and quartering seas.

3.1 Boundary Conditions

The first step of the simulation is to determine the forces acting on the catamaran, which are assumed to be concentrated on the sail for conservatism. From the standard drag formula [1], the force (F) can be related to a fluid's velocity (V) by the following equation:

$$F = 1/2 \cdot \rho \cdot C_D \cdot A \cdot V^2$$

Where, ρ is the density of the fluid (i.e., air at sea level), C_D is the drag coefficient, and A is the area of the sail. According to the ISO International Standard Atmosphere, the standard sea level density of air is approximately 1.225 kg/m³ [2]. Once again for conservatism, the wind is perfectly perpendicular to the sail, and, thus, the sail is assumed to be a long flat plate

perpendicular to flow. Given this, C_d is approximately 2, according to [3]. Finally, the sail is 10m^2 and V is 40 knots ($\approx 20.6 \text{ m/s}$):

$$F = 1/2 \times 1.225 \text{ kg/m}^3 \times 2 \times 10 \text{ m}^2 \times (20.6 \text{ m/s})^2 \approx 5,000\text{N}$$

With respect to the constraints, the mast is prevented from rotating freely by applying a glue-coincident connection between the mast and crossbeams in the .fem file. Afterwards, the two hulls are fixed.

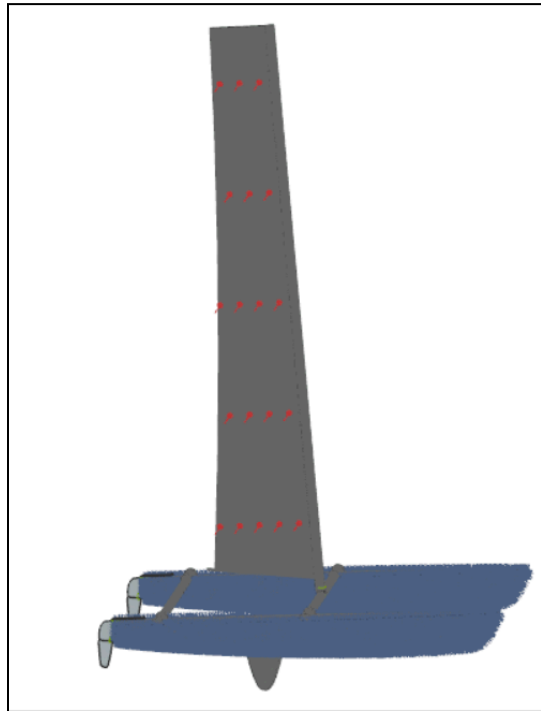
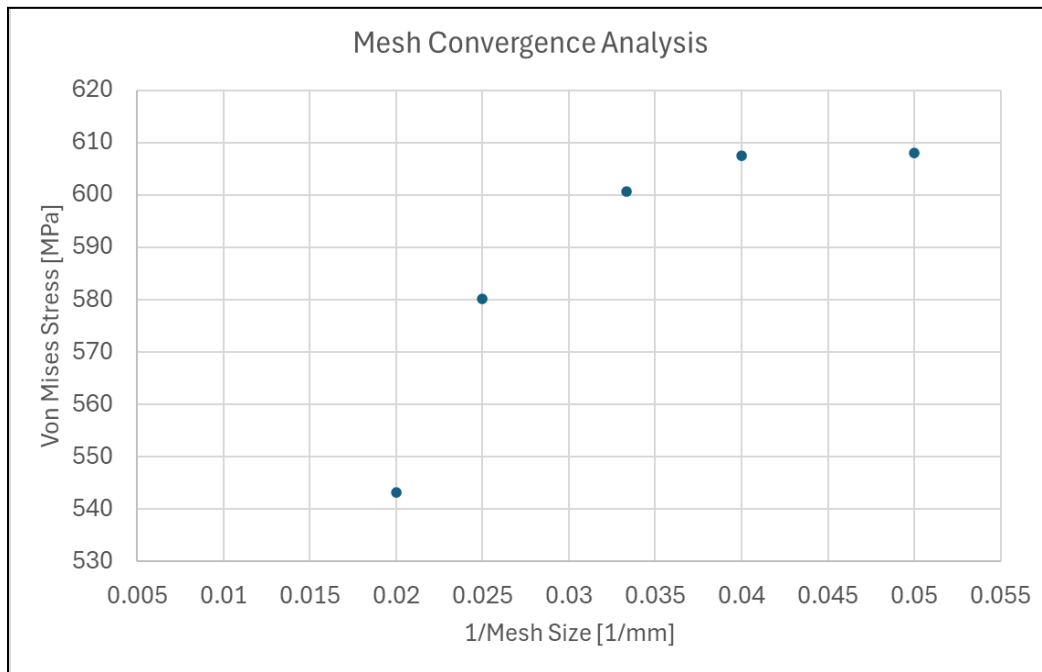


Figure: Forces and constraints for stress analysis during a fresh gale

3.2 Mesh Analysis

The mesh sizes were initialized at 50mm for the hulls and crossbeams, 20mm for rudders, and 50mm for mast and sail, using Swept Mesh for the sails and masts. Since the sail sees the most force and the mast the largest moment (near the connection with crossbeams), their meshes were gradually decreased until the maximum Von Mises elemental stress converged. If percent change in stress from one mesh size to the next was below 1%, then convergence was assumed.



Plot 1: Mesh convergence analysis for fresh gale simulation

3.3 Results

As seen in *plot 1*, the maximum Von Mises stress converged to 608.06 MPa at 20mm mesh size, which corresponded to a 0.08% change from the previous 25mm mesh size (607.55 MPa).

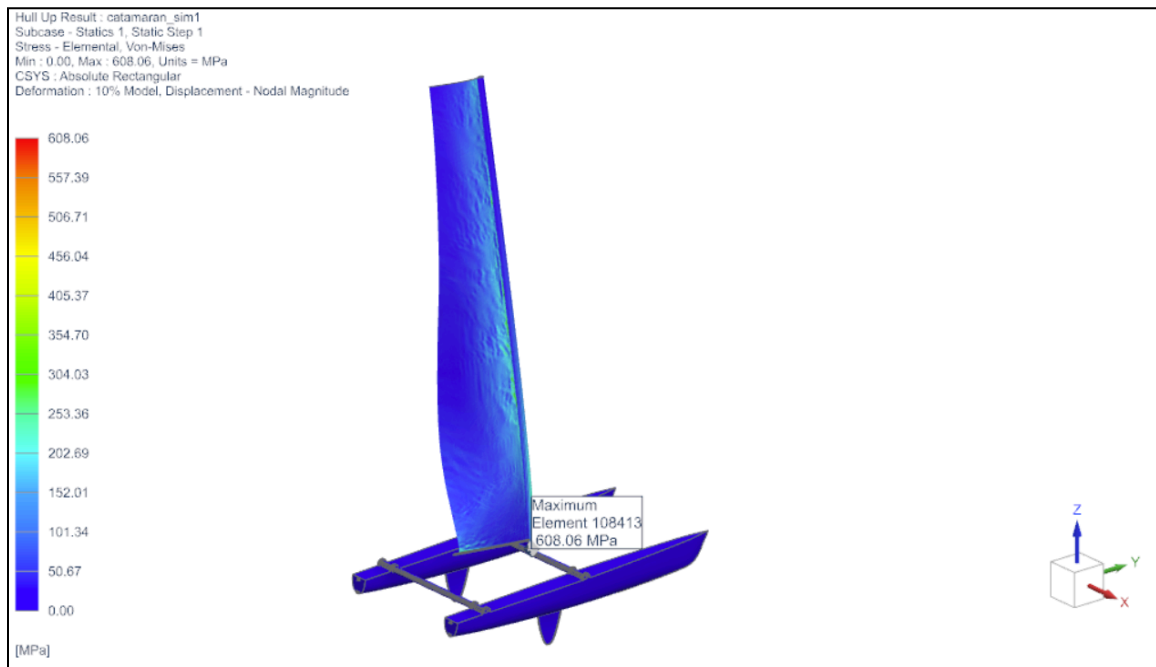


Figure: Von Mises elemental stress during a fresh Gale

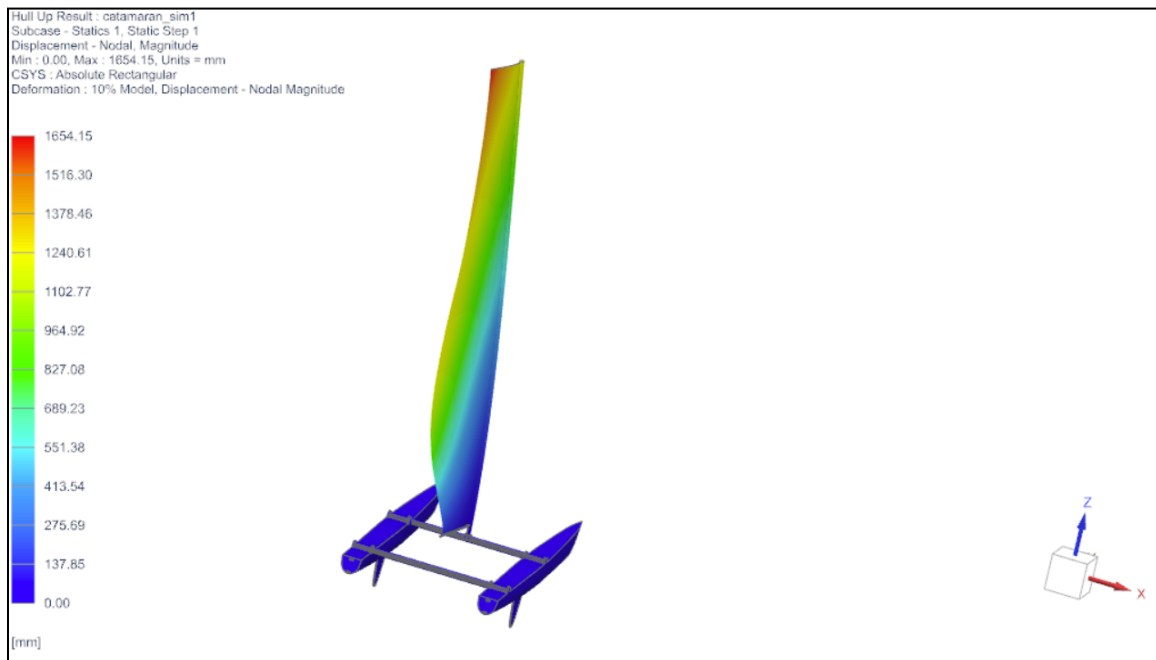


Figure: Displacement of mast and sail during a fresh Gale

The results above confirm engineering intuition which predicts the maximum displacement at the top of the mast and maximum stress near the bottom of the mast, which effectively acts as a vertical cantilever. Additionally, there are strong stress concentrations at the sail/mast interface, since the sailcloth is not allowed to freely conform its shape to the wind. On the other hand, there is a sharp decrease in stress in the sailcloth away from the mast because the sail can twist and bend with the wind – as confirmed by the Figure, in which the sailcloth takes on an “S”-like shape.

The max stress (608.06 MPa) is within the boundaries of reported yield strength for Epoxy-HS carbon fiber, resin infused woven fabric, biaxial lay-up (533 - 774 MPa). Since carbon fiber is highly dependent on the fabrication process and very sensitive to cracks and defects, it's advised to not sail during a Gale – especially since no safety factor was added to the result.

3.3.1 Validation

To validate the results for the stress analysis during a Gale, the mast was assumed to be a vertical cantilever of uniform circular cross-section and the wind forces (i.e., 5,000 N) were concentrated at the top of the mast. For even higher conservatism, the mast's diameter was assumed to be 75mm, which is the smallest diameter at the connection between the mast and the crossbeams. Consequently, the max stress estimation follows a simple beam formula in pure bending:

$$\sigma = \frac{M \cdot c}{I}$$

Where, M is the moment due to the wind force acting at the mast base, I is the second moment of area, and c is the distance from neutral axis to outer fiber (i.e., radius of mast). Therefore, after substituting in the equation for I and M , the theoretical max stress for the 9m mast is equal to:

$$\sigma_{Max} = \frac{32}{\pi} \cdot \frac{(F \times h)}{D^3} = \frac{32}{\pi} \cdot \frac{(5,000 \text{ N} \times 9000 \text{ mm})}{(75 \text{ mm})^3} = 1,086.5 \text{ MPa}$$

The hand calculations support the FEA's result, suggesting that it's structurally unsafe to sail in 40 knot winds. The International Tornado, being an Olympic boat designed for lightweight and speed, is not meant for rough weather conditions. Therefore, these results are not entirely surprising. Moreover, both the FEA and hand calculations introduce conservative assumptions, which increase the stresses on the vessel. These include completely fixing both hulls, which in reality are allowed to rotate and lift above the water, and ignoring the contribution of the daggerboards which resist lateral and rotational motion.

CHAPTER 4. FLYING A HULL STRESS ANALYSIS

Flying a hull is one of the defining operating conditions of a high-performance catamaran and a critical structural case for analysis. When the windward hull lifts clear of the water, the entire vessel effectively balances on a single, slender hull and a narrow hydrodynamic footprint. This dramatically alters the load paths and structural behavior becomes highly asymmetric. The leeward hull experiences increased bending and local slamming pressures, while the windward hull transitions to a suspended structural role, carrying tensile loads through the beams rather than buoyant support. Crossbeams see a combination of vertical shear, torsion, and in-plane

bending as they transfer the righting moment across the vessel. Appendages such as rudders and daggerboards must resist higher lateral forces as they provide directional stability at elevated speeds and reduced wetted area.

This chapter focuses specifically on the hulls and crossbeams to quantify stresses using linear static FEA, capturing how these elements deform and redistribute loads during sustained one-hull sailing. The goal is to determine whether the structural system maintains adequate stiffness, strength, and safety margins when operating at the performance limits for which the catamaran is designed.

4.1 Boundary Conditions

As aforementioned, the hull in contact with the water sees the most stress near the crossbeams. Here, the moment is largest when the opposite hull is barely above the water, since the moment arm is longest. Keeping with conservatism, the leeward hull is fully fixed, while the rest of the vessel is perfectly horizontal and gravity is acting on the entire structure.

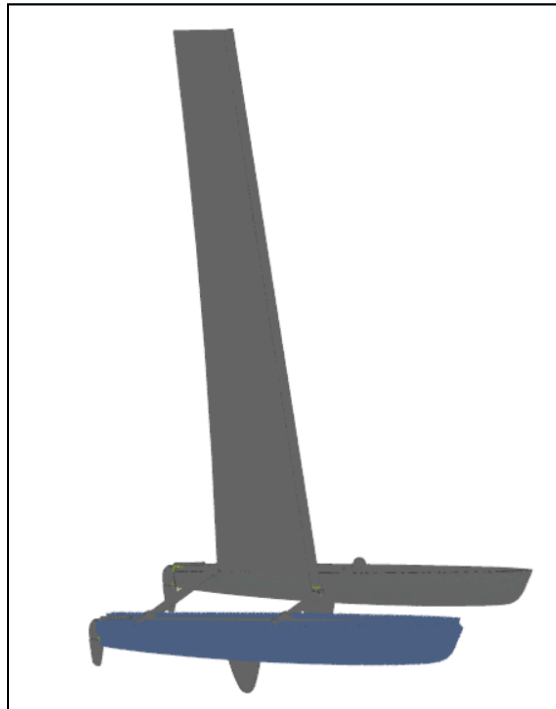
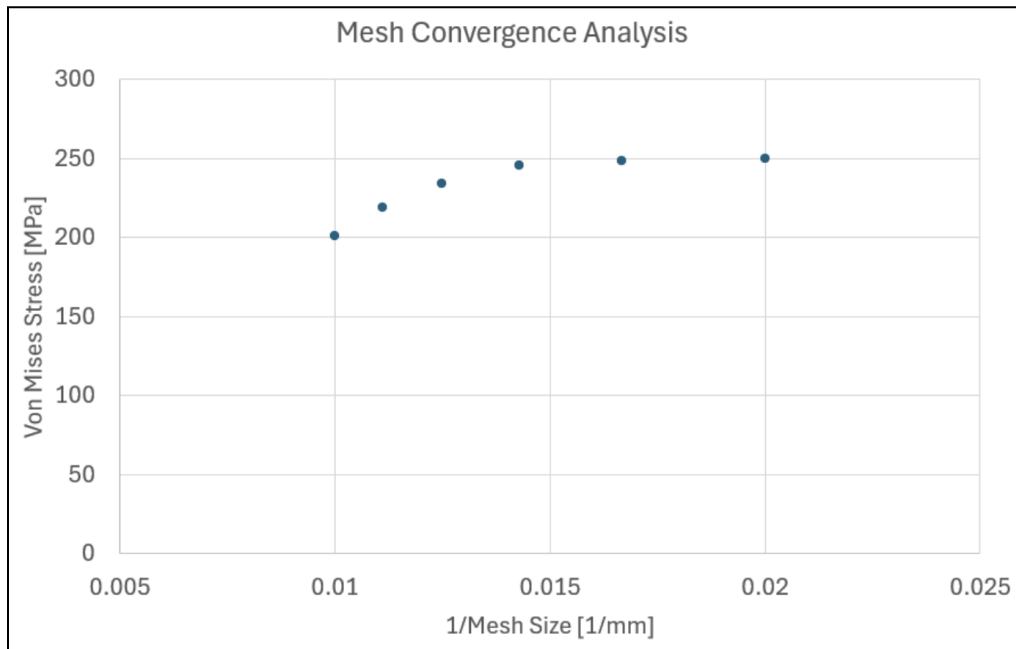


Figure: Constraints for flying a hull stress analysis. Siemens NX does not show gravity but acceleration is applied.

4.2 Mesh Analysis

The mesh sizes were initialized at 100mm for the hulls and crossbeams, and 20mm for rudders, mast and sail. Since the hulls and crossbeams see the largest stress, their meshes were gradually decreased until the maximum Von Mises elemental stress converged. Similarly to the previous analysis, if percent change in stress from one mesh size to the next was below 1%, then convergence was assumed.



Plot 2: Mesh convergence analysis for flying a hull simulation

4.3 Results

As seen in *plot 2*, the maximum Von Mises stress converged to 249.58 MPa at 50mm mesh size, which corresponded to a 0.55% change from the previous 60mm mesh size (248.22 MPa).

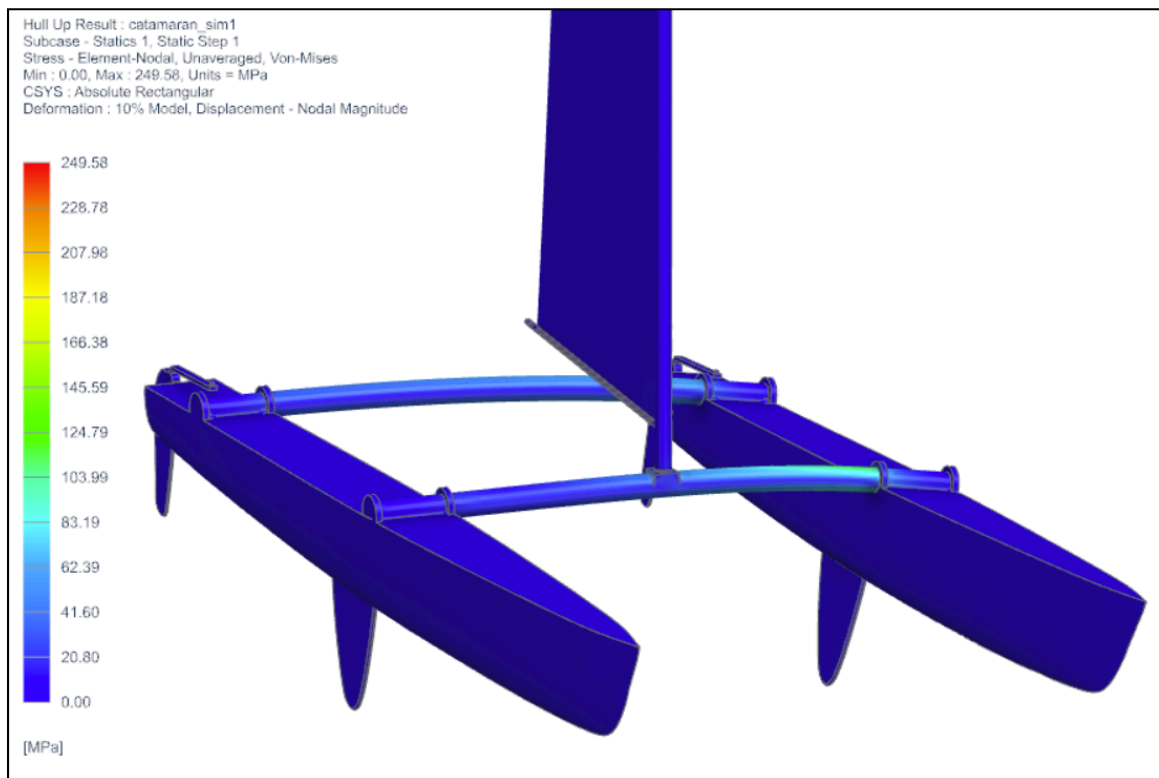


Figure: Von Mises elemental stress while flying a hull

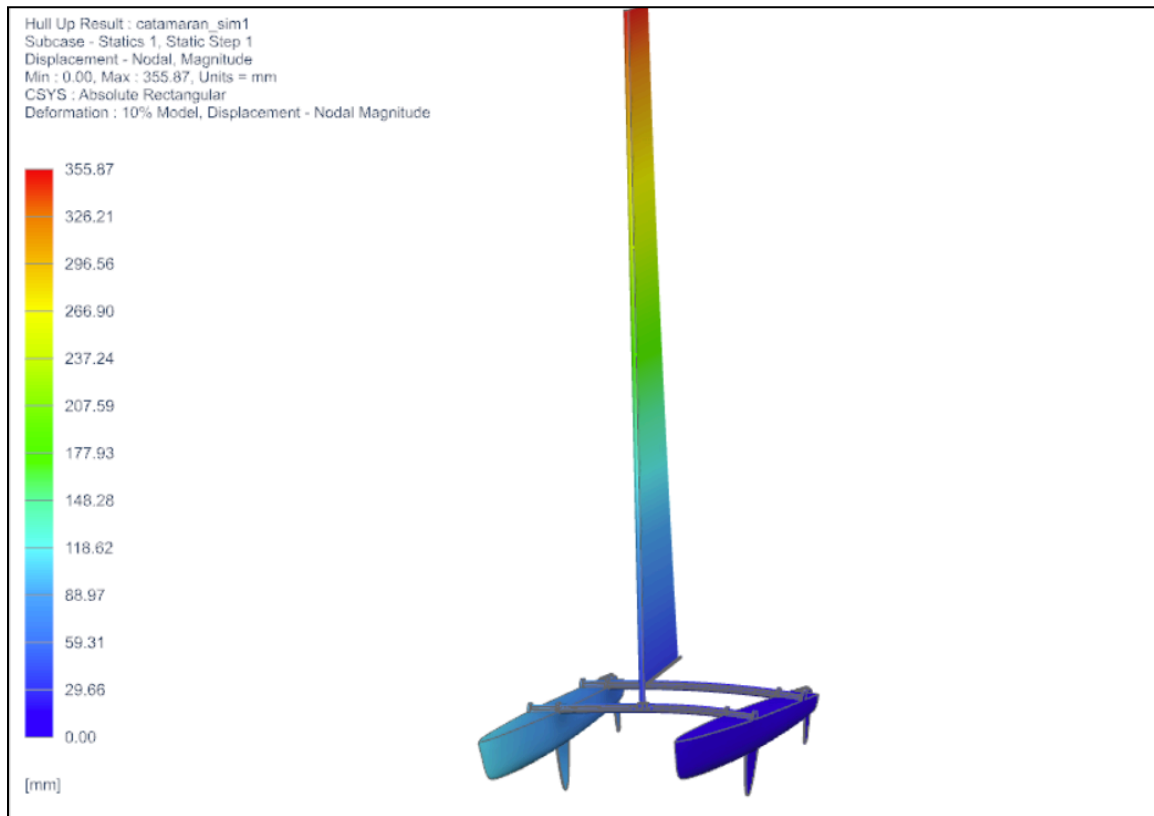


Figure: Displacement of mast and sail while flying a hull

The results above show that the maximum displacement occurs at the top of the sail and maximum stress near the crossbeams/hull joints. The max stress (249.58 MPa) is within the boundaries of reported yield strength for Epoxy-HS carbon fiber, resin infused woven fabric, quasi-isotropic lay-up (383 - 552 MPa). With a factor of safety of 2 (i.e., approximately 500 MPa), there is risk of failure, however, as highlighted above, the simulation includes highly conservative assumptions). Mainly, the vessel is not tilted and therefore the moment arm is considerably larger than it will ever be. Additionally, the contribution of the daggerboards is once again ignored, and the catamaran is allowed to deflect past the waterline.

4.3.1 Validation

To validate the FEA result, the crossbeams were modeled as a simple cantilever, with the weights of the beams themselves, the mast and sails, and flying hull as loads. The masses of the hull, mast and sails, and crossbeams were found by multiplying their volume by density. The volumes and distance from the fixed end of the cantilever beam were found by using the measure tool in NX.

	Volume (m^3)	Density (kg/m^3)	Weight (N)	Moment Arm (mm)	Moment (Nmm)
Hull	0.8419	1515	12513	2450	30655837
Beams	0.0355	1515	527.63	1115	588306
Mast	0.0890	1525	1332.02	1090	1451905
Total bending moment					32696048

The bending moment of inertia of the beam is:

$$I = \frac{\pi}{64} (d_1^4 - d_2^4) = \frac{\pi}{64} (125^2 - 75^2) = 10.4 \times 10^6 \text{ mm}^4$$

Where d_1 and d_2 are the outer and inner diameters of the crossbeam.

Thus the max stress using this simple cantilever model is:

$$\sigma = \frac{Md_1}{2I} = \frac{(32696048)(125)}{(2)(10431069)} = 195.91 \text{ MPa}$$

The simple cantilever model yields a max stress of around 196 MPa, while the FEA simulation yields a max stress of around 250 MPa. The FEA predicts a max von Mises stress

that is about 28% higher, which is reasonable as they are on about the same order of magnitude. The difference is likely due to local stress concentrations at the beam-to-hull connection and the simplified assumption of a single cantilever beam rather than the full two-beam, torsion bearing structure.

CHAPTER 5. DRAFT ANGLE OPTIMIZATION (CFD)

5.1 Geometry and Computational Domain

The starting point for the CFD work was the full catamaran assembly (two hulls, cross-beams, and rudders). The model was first defeatured, removing parts that are irrelevant for the large-scale flow (footholds, small fillets, fasteners, cables, etc.). This produced a clean “hydrodynamic” geometry while keeping all shapes that influence pressure and shear on the hulls and rudders.

To create the fluid domain, a large rectangular block of water and a second block of air were modeled around the boat. The hulls were then subtracted from these solids so that the remaining volumes represented only fluid. This Boolean subtraction ensures that the hull surfaces appear as wetted boundaries in the flow model and remain perfectly consistent with the CAD used for structural design.

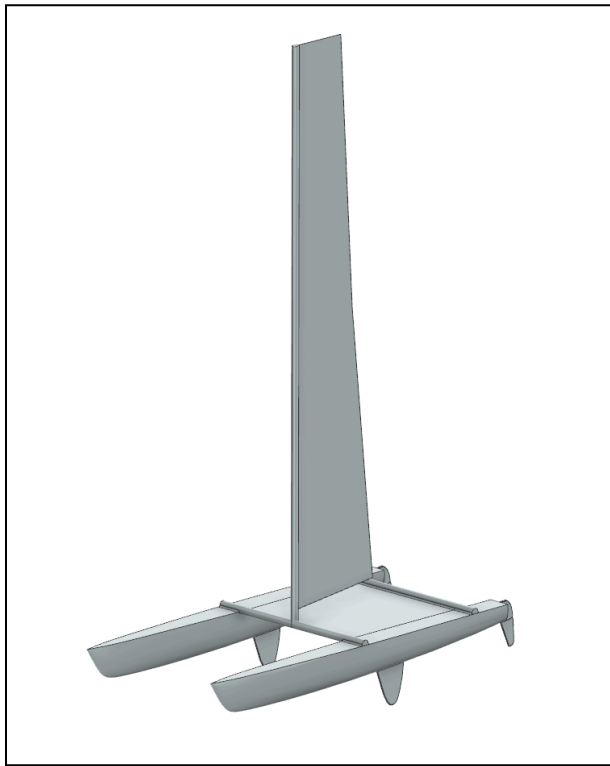


Figure: Simplified Catamaran Geometry

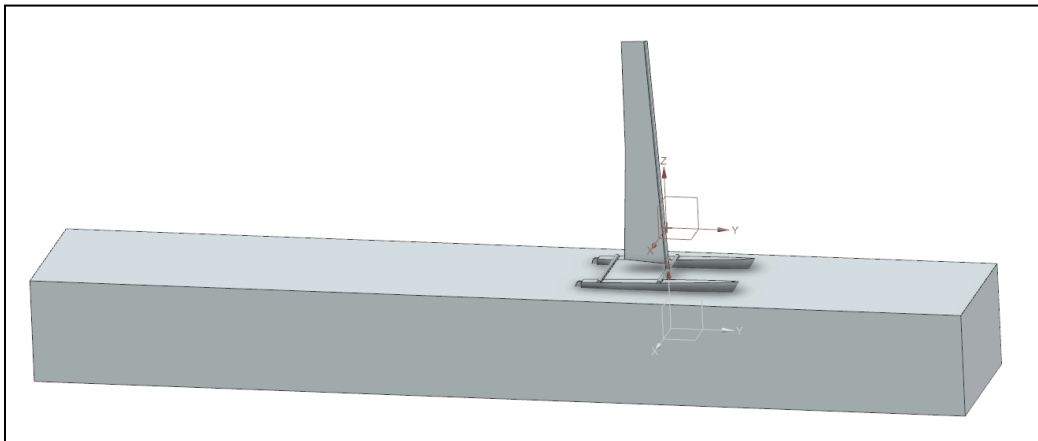


Figure: Solid subtraction process for liquid domain

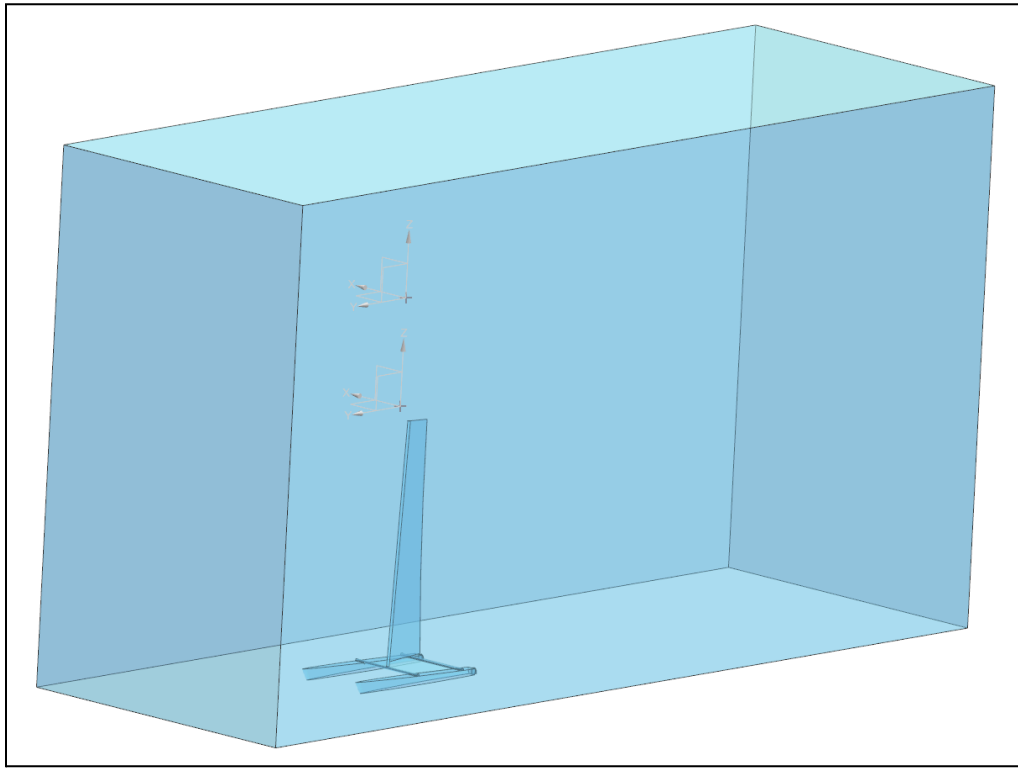


Figure: Subtracted solid for air domain

For the both simulations, the block was sized following standard external-aerodynamics guidelines:

- inlet plane located about 3 hull lengths upstream of the bow
- outlet plane located about 4 hull lengths downstream of the stern
- lateral and vertical boundaries at least 2–3 hull widths away from the hulls

These distances follow typical external-aerodynamics guidelines so that the outer boundaries behave approximately as far-field and do not artificially accelerate or decelerate the flow around the boat. For the water simulation, only the volume below the still-water line was kept; for the air simulation, the “water surface” plane was treated as a solid wall and only the portion of the hulls

above the water line remained in contact with air. The two fluids were therefore modeled in separate simulations rather than with a full free-surface model.

5.2 Hull Immersion and Test Matrix

The static draft of the catamaran (boat at rest) is approximately 0.127 m, obtained from a simple buoyancy balance:

$$\text{Boat weight: } W = m_{\text{boat}} \cdot g.$$

$$\text{Displaced water weight: } W = \rho_{\text{water}} \cdot g \cdot V_{\text{disp}}$$

Given the hull waterplane area (from NX Measure tool), solving for the submerged volume gives a still-water immersion of about 0.127 m.

In operation, hydrodynamic lift acts on the hulls and effectively reduces the immersed depth (“dynamic squat”) as speed increases. Instead of solving a full free-surface problem, we approximated this effect with a simple empirical-style rule¹ that prescribes immersion as a function of boat speed.

V (km/h)	d(V) (m)
20	0.108
30	0.094
40	0.079
50	0.065

¹ He, W., Castiglione, T., Kandasamy, M., & Stern, F. “Numerical analysis of the interference effects on resistance, sinkage and trim of a fast catamaran.” *Journal of Marine Science and Technology*, 20, 292–308 (2015).

60	0.051
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Table: Velocity Parametric Study Water Immersion Values

For each operating point between 20 and 60 km/h, the hulls were translated vertically so that this immersion depth lay below the still-water plane before the fluid volumes (water and air) were created and meshed. In other words, we “baked in” dynamic squat by positioning the hulls, rather than by explicitly modeling the time-varying free surface.

5.3 Meshing Strategy

The water / air volume was discretized with 3D tetrahedral elements (TET4). The initial mesh used a relatively coarse global element size to keep the problem small, but with refinement regions wrapped around the hulls and rudders:

- smaller surface elements on hulls, beams, and rudders;
- tighter volume growth in the near-wall region to better capture boundary layers and wakes;
- coarser elements in the far-field, where gradients are weaker.

Mesh refinement was carried out iteratively:

1. Run a case with a coarse mesh and record drag and lift coefficients.
2. Refine the mesh near the hulls (reduced element size and growth rate).
3. Re-run and compare forces, moments, and coefficient histories.
4. Repeat until the change in drag and lift coefficients between successive meshes was only a few percent, smaller than other modeling uncertainties.

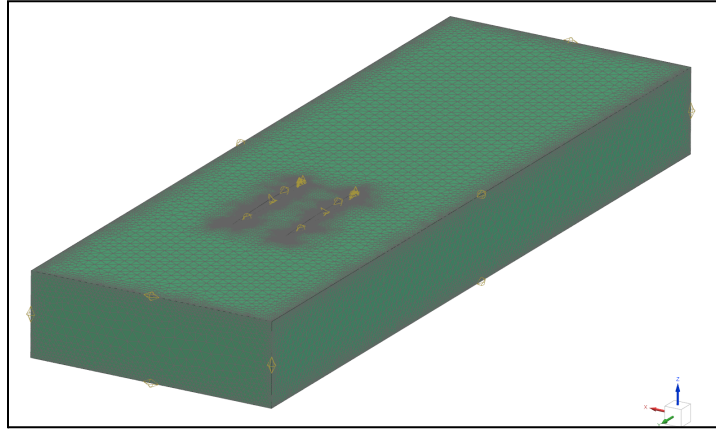


Figure: Final mesh for water domain

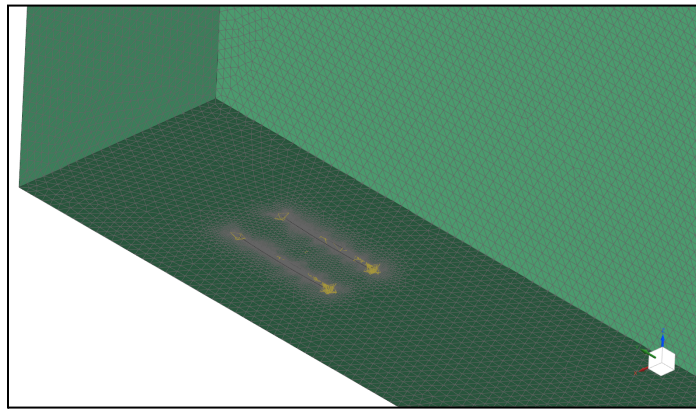


Figure: Final mesh for air domain

The final water mesh for the two-hull configuration contained roughly 4.9×10^5 volume elements and about 1.1×10^5 nodes, with the densest elements on the hulls, beams, and rudders. The same meshing strategy was used for the air domain. These final meshes were then reused across the entire speed/immersion matrix parametric study for consistency.

5.4 Boundary Conditions and Solver Configurations

All simulations used the Advanced Flow solver in Simcenter 3D Thermal/Flow with:

- Steady-state, incompressible, isothermal water at 20 °C (and air at 20 °C in the air cases).
- Density and viscosity taken from the built-in “Water[1]” and “Air” material definitions.
- Gravity acting in the negative z-direction. Buoyancy modeling itself was turned off; the hydrostatic effect is represented through the prescribed immersion depths rather than a free surface.
- Turbulence model: SST (shear stress transport) two-equation RANS model.

Initial trials used first-order advection, which is robust but smears velocity and pressure gradients, artificially thickens wakes and boundary layers, and can under-predict suction peaks that contribute to lift and drag. To reduce this numerical diffusion, the production runs used Second Order (CDS) for energy (kept consistent even though the flow is nominally isothermal) and turbulence quantities (k and ω). Limiter values of 0.75 for momentum/energy and 0.25 for turbulence were applied. These limiters prevent non-physical (numerical) oscillations while still preserving a mostly second-order distribution of the quantities. The practical effect is a better wake, more accurate pressure distribution over the hull and more reliable integrated drag and lift.

In a steady-state RANS calculation, the “physical time step” controls how aggressively the solver updates the flow field each global iteration. Early runs used a time step of 0.5 s. After upgrading to second-order schemes, the turbulence and energy equations became stiffer, and the high-order discretization left less numerical damping. To keep the nonlinear iterations stable and improve convergence of the turbulence residuals, the time step was reduced to 0.1 s.

At the same time, relaxation factors were tuned for stable but reasonably fast convergence, including global relaxation factor: 0.7, mass relaxation factor: 0.75, fluids relaxation factor: 0.9, turbulence relaxation factor: 0.9.

The key boundary conditions for the water simulations were:

- Inlet: velocity inlet with uniform profile aligned with the boat x-axis. For example, at 20 km/h: $V = 20 \text{ km/h}$. The other test velocities were 30, 40, 50 and 60 km/h.
- Outlet: “Opening” boundary with specified absolute pressure and temperature: $p = 1.013 \times 10^2 \text{ mN/mm}^2$ (1 atm), $T = 20^\circ\text{C}$. This behaves like a pressure outlet while allowing limited backflow if the wake expands.
- Hulls, rudders, beams: no-slip walls with zero roughness. The default forced-convection heat transfer was left on (thermal effects are negligible but this keeps the model options consistent).
- Far-field boundaries (top, sides, bottom): walls placed far enough away that the flow near the hulls sees them as approximately free-stream.

For each run, a Lift and Drag Report, specifying the manifold equivalent area, length and approach velocity, was attached to all wetted (by water or air) hull, rudder and mast surfaces to obtain integrated forces, moments, and non-dimensional coefficients (drag, lift, side force, and pitching/yawing moments).

5.5 Convergence Monitoring

Convergence and validity of solution were evaluated using three complementary criteria:

- Residuals. RMS residuals for momentum and continuity decreased several orders of magnitude, typically stabilizing in the 10^{-4} – 10^{-5} range by the end of the run. Turbulence residuals (k and ω) converged more slowly (common in steady RANS), but continued to decrease once the smaller time step and second-order schemes were in place.
- Global balances. The solver's summary reported final mass and momentum imbalances well below 1 %. For example, the final 50 km/h two-hull water run had a mass imbalance of order 10^{-5} % and momentum imbalance of order 10^{-3} %. This indicates a nearly perfect balance between the inflow, outflow, and reaction forces on the hulls.
- Integral quantities. Drag and lift coefficients from the Lift/Drag report were monitored as functions of iteration. Iterations were continued until these curves reached a flat plateau with negligible change between successive iterations. Even when turbulence-energy imbalance was still on the order of tens of percent, the integrated forces had already stabilized, which is a standard acceptance criterion for steady RANS studies.

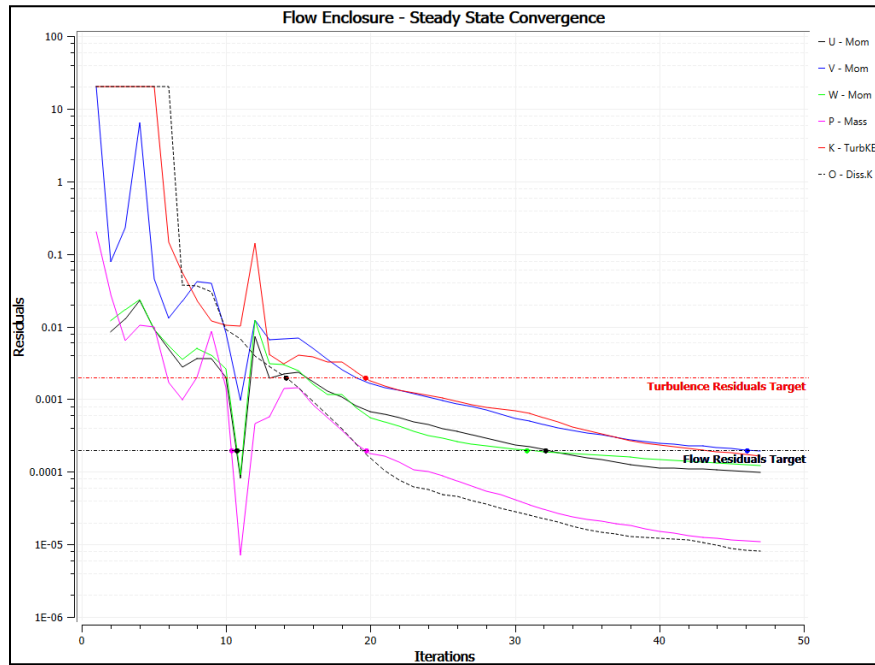


Figure: Residuals vs. Iterations for water domain at 60 km/h

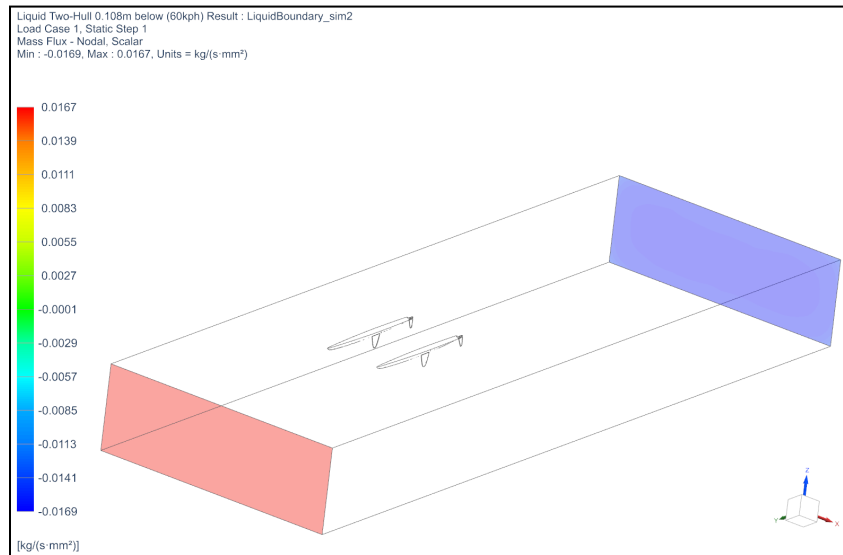


Figure: Mass Flux Balance at Inlet and Outlet for 60 km/h water (as seen, >1% difference)

Only runs satisfying all three criteria were retained in the final data set.

5.6 Flow Field Characteristics

Post-processing of the two-hull water case showed flow features consistent with expectations for a streamlined, semi-buoyant body at high Reynolds number. Additionally, since validation of fluid flow simulations is nearly impossible to be performed without physical experimentation (especially not with hand calculations), this section attempts to support the results with theoretical explanations for them.

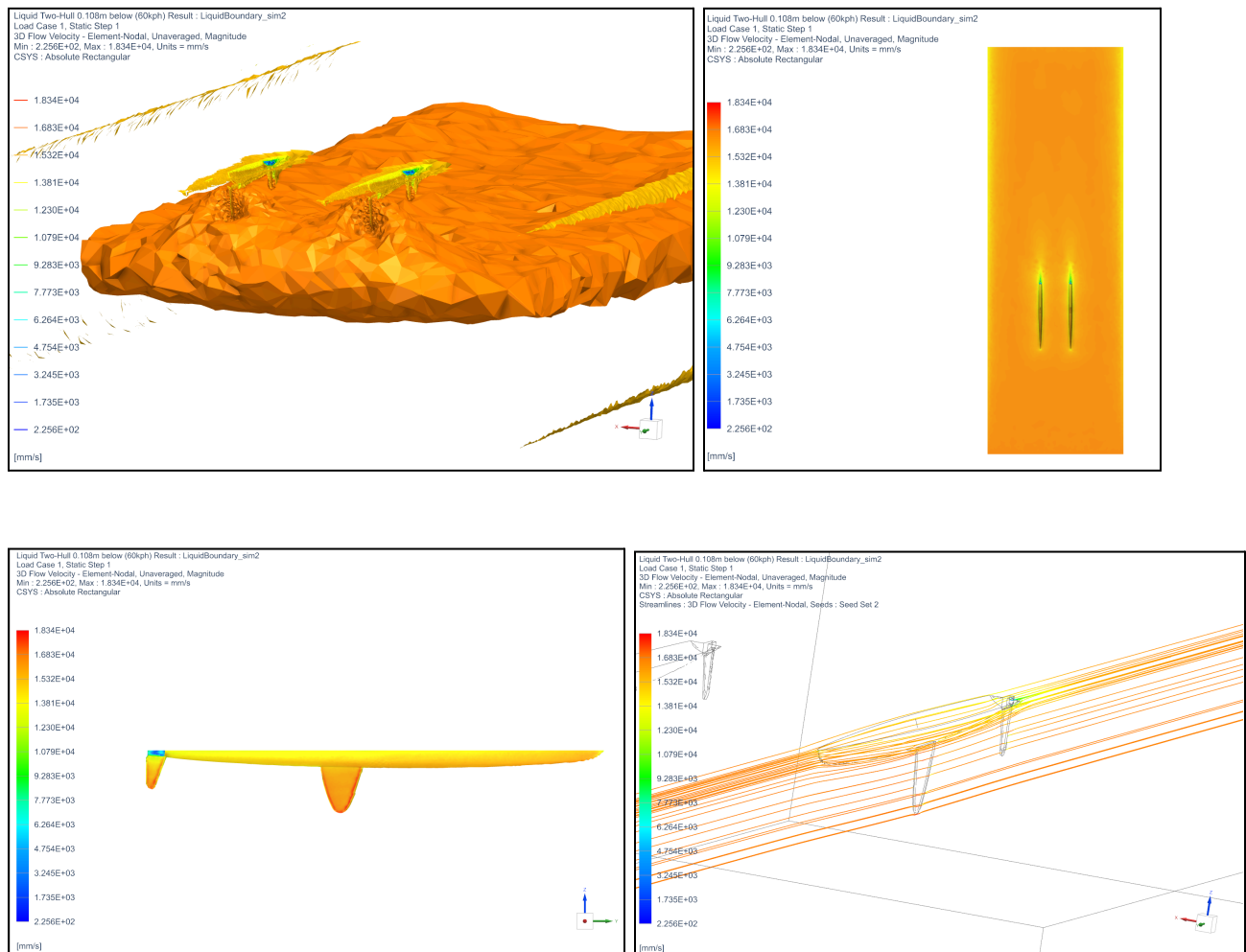


Figure: Selected Velocity Plots for Water Domain

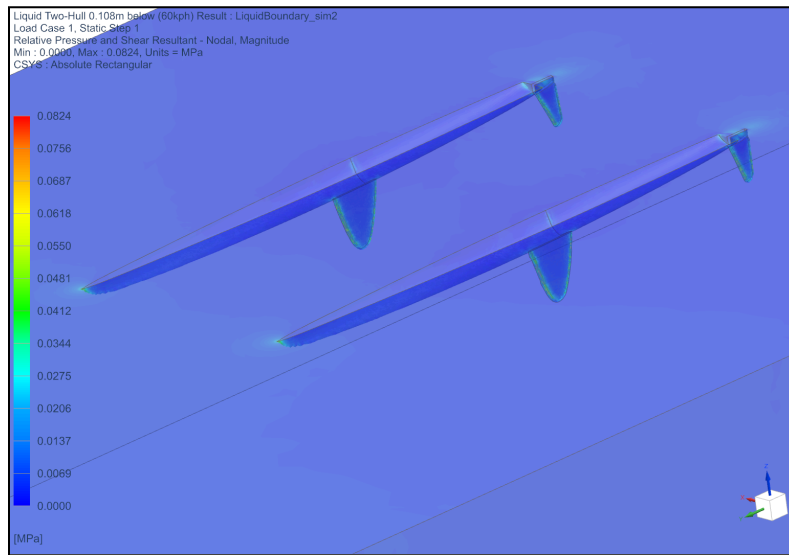


Figure: Relative Pressure and Shear Resultant Plot

As can be seen, velocity magnitude drops to nearly zero at the nose of each hull, with a corresponding local maximum in static pressure. This is the stagnation point, where dynamic pressure is fully converted into static pressure. As water is guided around the hulls, the effective cross-sectional area available to the flow decreases and streamlines curve around the bow. Velocity increases along the shoulders and mid-length of each hull, with a slight drop in static pressure relative to the free stream (Bernoulli effect). This pressure deficit contributes to both drag and hydrodynamic lift. Each rudder, held at zero steering angle, shows low velocity along its leading edge and a roughly symmetric velocity distribution on either side. Velocity increases over the surfaces toward the trailing edge, with only small separation regions near the tip, consistent with a thin foil at small angle of attack. Downstream of the hulls and rudders, a low-velocity region develops that gradually recovers toward the inlet velocity. The pressure field mirrors the velocity field: high pressure at the bow, low pressure along the shoulders, and a gradual recovery toward ambient in the far wake.

These patterns confirm that the mesh and turbulence model are capturing the main physics of drag and lift generation for this hull shape.

5.7 Parametric Study and Trends

V (km/h)	V (m/s)	Draft d (m)	Drag F_D (N)	C_D	Lift F_L (N)	C_L	P_{max} (mN/mm²)	P_{min} (mN/mm²)	P_{avg} (mN/mm²)
20	5.56	0.108	716	135.98	-669	127.0	9.16	-8.77	0.278
30	8.33	0.094	1398	117.97	-1306	110.2	20.6	-19.7	0.626
40	11.11	0.079	2106	99.97	-1967	93.4	36.6	-35.1	1.11
50	13.89	0.065	2698	81.97	-2520	76.6	57.2	-54.8	1.74
60	16.67	0.051	3032	63.96	-2831	59.7	82.4	-79.0	2.50

Table: Parametric Study for Water Domain

The same setup (geometry, domain, mesh, solver options) was used across the matrix of speeds from 20 to 60 km/h and corresponding immersion depths from 0.108 m to 0.0508 m. For each operating point, the Lift/Drag report returned drag force F_D , drag coefficient C_D , vertical force F_L , lift coefficient C_L , and several pressure metrics.

The drag force increases strongly with speed: from 716 N at 20 km/h to about 3030 N at 60 km/h, which is broadly consistent with the expected quadratic scaling:

$$F_D \propto \frac{1}{2} \rho V^2 C_D A_{ref}$$

If C_D and reference area were constant, F_D would increase exactly with V^2 ; in practice, the increase is somewhat weaker because the immersed volume (and effective wetted area) decreases with speed.

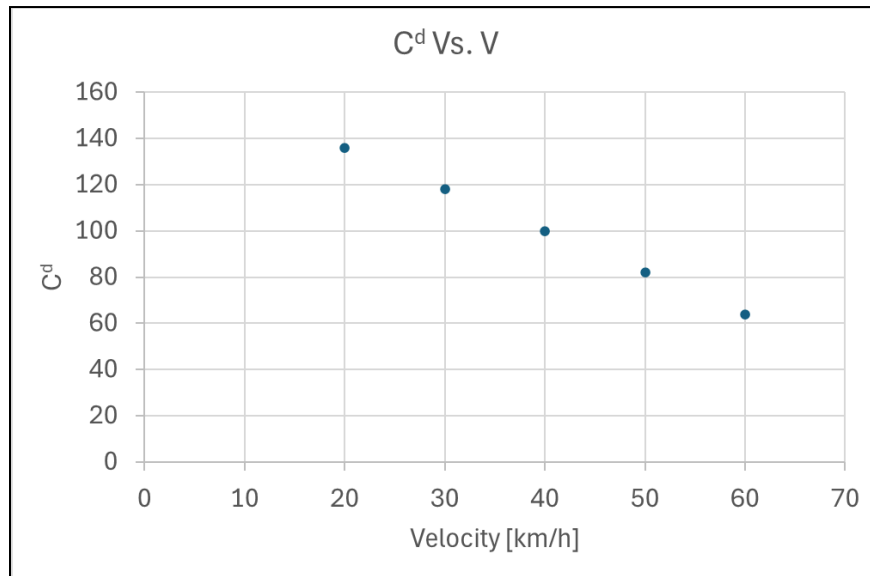


Figure: Drag coefficient vs. speed (C_D vs V).

The drag coefficient itself decreases from about 136 at 20 km/h to about 64 at 60 km/h. Since C_D is defined as:

$$C_D = 2 \frac{F_D}{\rho V^2 A_{ref}}$$

This drop means that, relative to the dynamic pressure, the hull becomes aerodynamically “cleaner” as it rides higher in the water. Physically, less hull surface is inside the water, so the cross-sectional area facing the flow is reduced, and the flow remains attached over a larger

fraction of the hull. In other words, part of the apparent speed dependence of C_D is really a geometry change due to reduced immersion.

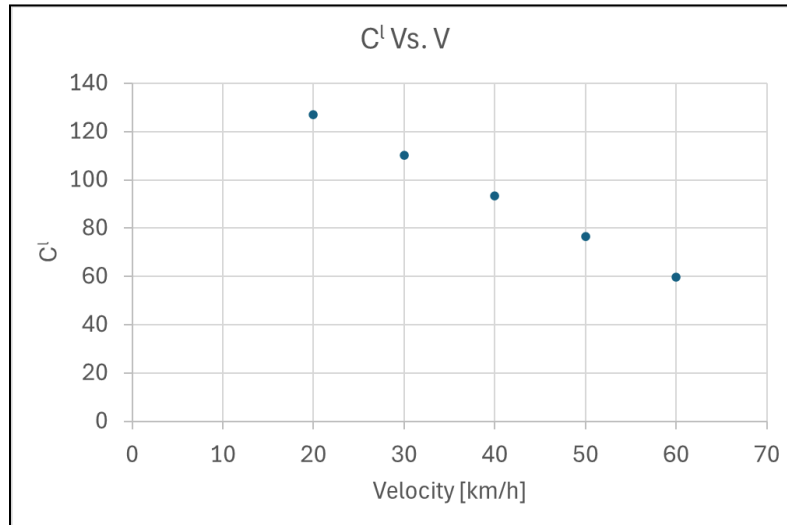


Figure: Lift coefficient vs. speed (C_L vs V).

The vertical hydrodynamic force magnitude $|F_L|$ grows with speed, from about 670 N at 20 km/h to about 2830 N at 60 km/h. This reflects the increase in dynamic pressure and the larger pressure differences between the sides and underside of the hulls at higher speeds.

However, the lift coefficient, $C_L = 2 \frac{F_D}{\rho V^2 A_{ref}}$, decreases from roughly 127 to 60 over the same range. This mirrors the behavior of C_D and again indicates that, measured relative to dynamic pressure, the hull is generating less lift per unit area at higher speeds because less of it is in the water. The hulls are effectively transitioning toward a semi-planing state: absolute lift still increases with speed and supports the boat, but the immersion and effective “angle of attack” of the underwater volume are reduced.

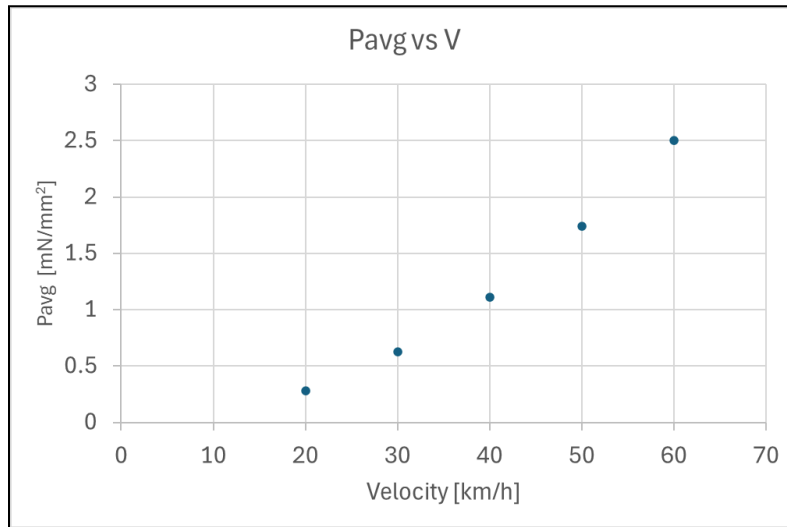


Figure: Average pressure vs. speed (P_{avg} vs V).

The maximum and minimum static pressures on the hull surfaces increase in magnitude with speed (from roughly ± 9 mN/mm² at 20 km/h to about +82, −79 mN/mm² at 60 km/h). This is consistent with dynamic pressure: as V increases, local stagnation pressures scale with $\frac{1}{2}\rho V^2$.

The average static pressure on the hulls, P_{avg} , increases from about 0.278 mN/mm² to 2.50 mN/mm². This does not mean the flow becomes more hydrostatic. Rather, it shows that a larger fraction of the hull is exposed to bigger pressure gradients at higher speeds.

Together, these plots show that absolute drag grows roughly as V^2 , as expected for turbulent external flow, but is significantly mitigated by reduced immersion at higher speed. Drag and lift coefficients decrease across the range because the hull geometry seen by the water changes (less wetted area and more favorable flow alignment as the boat rides higher). Pressure extremes scale with dynamic pressure, confirming that the model captures the expected stagnation / suction behavior around the hulls and rudders. Dynamic lift is strong enough that, if

the trend continued beyond 60 km/h, further reductions in immersion would be expected, though not normal with both hulls in the water, consistent with the known tendency of racing catamarans to “fly” a hull at high speed. This all provides qualitative and quantitative evidence that the hull can generate enough lift to reduce immersion and drag, while also showing how sensitive resistance is to relatively small changes in ride height.

V (km/h)	V (m/s)	Umax (m/s)	Uavg (m/s)	Umin (m/s)
20	5.56	6.69	5.26	0.056
30	8.33	10.04	7.89	0.084
40	11.11	13.39	10.52	0.112
50	13.89	16.74	13.15	0.093
60	16.67	20.08	15.78	0.053

Table: Parametric Study on Velocity for Air Domain

For the air domain, similar trends were seen. For the 60 km/h air case ($V = 16.67$ m/s, draft $d = 0.0508$ m), the solver reached steady convergence in about 40 global iterations. Residuals for momentum and continuity dropped to the 10^{-4} level, turbulence residuals continued to decay, and global mass and momentum imbalances were of order 10^{-5} % and 10^{-3} %, respectively, similar to the water runs. Pressure, velocity, and turbulence monitors all flattened out, indicating a statistically steady solution.

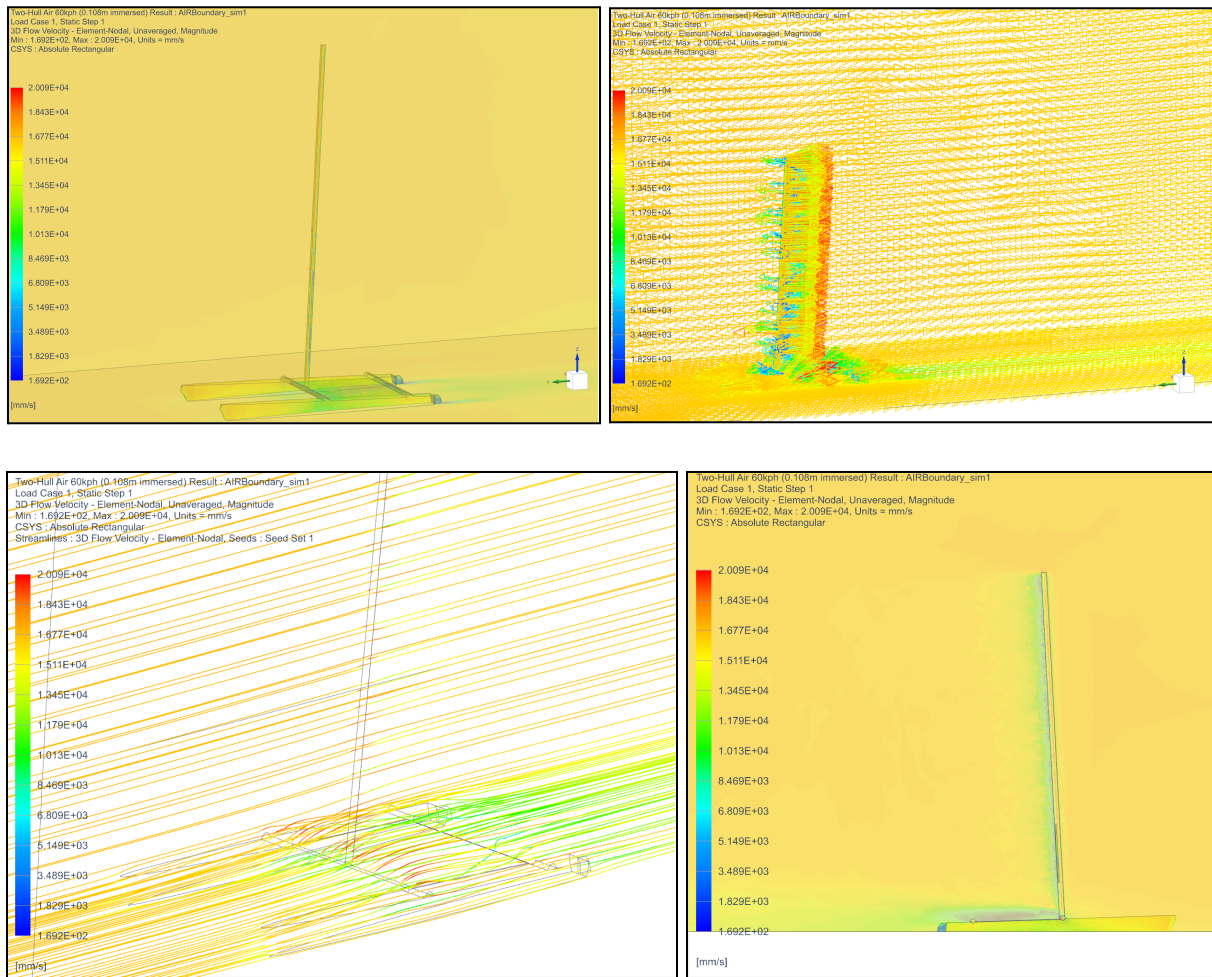


Figure: Selected Velocity Plots for Air Domain

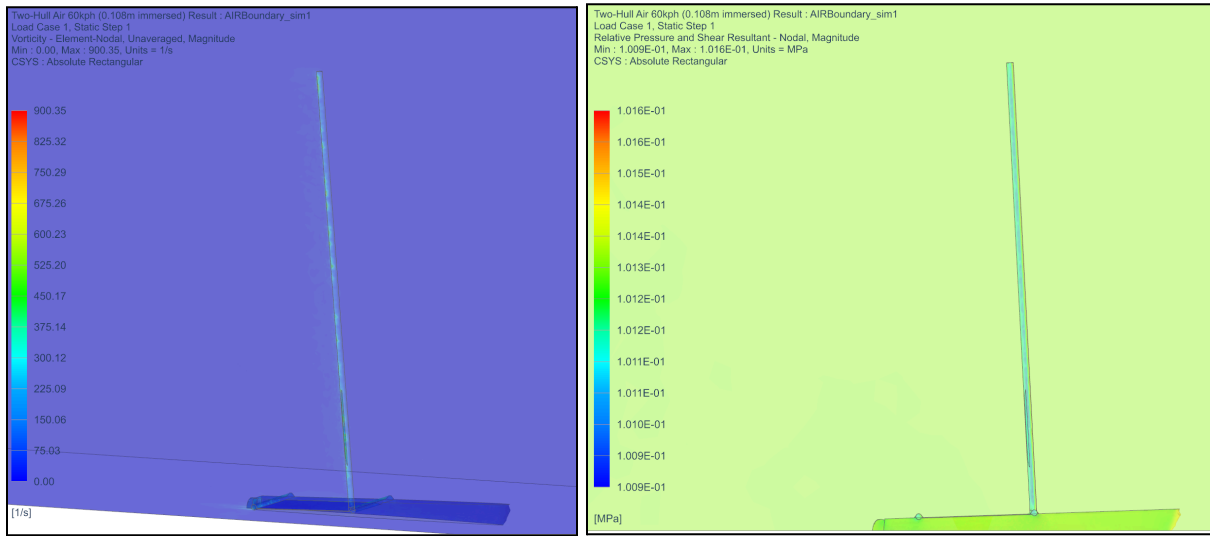


Figure: Vorticity and Relative Pressure Plots for Air Domain

The velocity and pressure fields show the expected external-aerodynamic features for a streamlined hull with attached appendages. A stagnation region forms at the bow of each hull above the waterline, at the leading edge of the cross-beam, and on the windward side of the mast. Local velocity goes to nearly zero at these points and static pressure peaks slightly above ambient. Around the rounded bow and cylindrical mast, the flow accelerates; the 60 km/h case yields a maximum velocity $U_{\max} \approx 20.1$ m/s compared with the inlet $U \approx 16.7$ m/s (about 20 % acceleration). This is consistent with streamlines being squeezed around the hull and mast cross-sections. On the leeward sides of the mast and hull tops, the flow slows down and small separation bubbles appear, followed by a wake with reduced velocity ($U_{\min} \approx 0.05$ m/s) and increased turbulence kinetic energy. Downstream of the rig, the wake gradually recovers toward the inlet velocity and pressure, with no evidence of large-scale unsteadiness or recirculation that would compromise the steady RANS assumption.

Pressure variations in air are much smaller in absolute terms than in water because the dynamic pressure is several orders of magnitude lower. At 60 km/h the solution gives $P_{\max} \approx 101.64 \text{ mN/mm}^2$ and $P_{\min} \approx 100.96 \text{ mN/mm}^2$, i.e., excursions of only a few tenths of a kPa around ambient ($\approx 101.3 \text{ kPa}$). These values are of the same order as the theoretical dynamic pressure of air at 16.7 m/s, so the air solution is self-consistent. Integrated over all above-water surfaces, this pressure distribution produces a drag force $F_D \approx 98.8 \text{ N}$ and a small net vertical force $F_L \approx -30.3 \text{ N}$ (negative sign following the solver's convention, corresponding to a downward load on the hull).

For comparison, at 60 km/h the hydrodynamic drag from the water solution is about 3030 N and the hydrodynamic vertical force magnitude is about 2830 N. Thus, air drag is only about 3% of total drag at this speed, and the aerodynamic vertical force is roughly 1% of the hydrodynamic vertical support, again, reinforcing the advantage of reducing the water wetted area. With this, aerodynamic loads are secondary for straight-line running, but they are not completely negligible for rig design and dynamic stability.

The air simulations were repeated for the same speed/draft combinations used in the water study, giving the following force and pressure trends.

V (km/h)	V (m/s)	Draft d (m)	Drag F_D (N)	C_D	Lift F_L (N)	C_L	Pmax (mN/ mm ²)	Pmin (mN/ mm ²)	Pavg (mN/ mm ²)
20	5.56	0.1080	11.9	0.95	-2.5	0.20	101.34	101.24	101.30
30	8.33	0.0940	25.8	0.92	-6.5	0.23	101.39	101.21	101.30
40	11.11	0.0790	44.9	0.90	-12.5	0.25	101.45	101.15	101.30

50	13.89	0.0650	69.4	0.89	-20.3	0.26	101.59	101.06	101.30
60	16.67	0.0508	98.8	0.88	-30.3	0.26	101.64	100.96	101.30

Table: Parametric Study for Air Domain

At 20 km/h ($V = 5.56$ m/s, $d = 0.108$ m) the above-water surfaces experience $F_D \approx 11.9$ N and $F_L \approx -2.5$ N, with drag coefficient $C_D \approx 0.95$ and lift coefficient $C_L \approx 0.20$. At 30, 40, 50, and 60 km/h the drag rises to 25.8, 44.9, 69.4, and 98.8 N, respectively, while C_D decreases only slightly from 0.95 to 0.88. Over the same range, the vertical force magnitude grows from about 2.5 N to 30.3 N and C_L increases from roughly 0.20 to about 0.26. The maximum and minimum surface pressures follow the same trend: $P_{\max}$ increases from ≈ 101.34 to ≈ 101.64 mN/mm², $P_{\min}$ decreases from ≈ 101.24 to ≈ 100.96 mN/mm², while the area-averaged pressure P_{avg} remains essentially constant at ≈ 101.30 mN/mm² for all speeds.

These data can be interpreted as follows:

- Drag force vs. speed (F_D vs V). Aerodynamic drag in air grows strongly with speed, from ~ 12 N at 20 km/h to ~ 99 N at 60 km/h. Over this range, V increases by a factor of 3 and V^2 increases by a factor of 9; the drag increase is slightly less than quadratic because C_D decreases modestly with Reynolds number. This is consistent with the standard drag law $F = 1/2 \cdot \rho \cdot C_D \cdot A \cdot V^2$, with geometry fixed and only weak Reynolds-number dependence of C_D .
- Drag coefficient vs. speed (C_D vs V). Unlike the water case, where immersion changes the effective geometry, the above-water geometry seen by the air is nearly constant. As a result, C_D is almost flat: it drops gently from about 0.95 at 20 km/h to about 0.88 at 60

km/h. This small reduction is typical of bluff bodies at increasing Reynolds number: the boundary layer becomes fully turbulent, separation points shift slightly downstream, and pressure drag is reduced a little.

- Lift coefficient vs. speed (C_L vs V). The vertical force magnitude increases from ~ 2.5 N to ~ 30.3 N as speed increases, and C_L rises from about 0.20 to ~ 0.26 . This indicates that, relative to dynamic pressure, the above-water shape becomes slightly more “lifting” (here, producing a small downward force on the hull) at higher speeds. Physically, stronger suction on the leeward sides of the mast and hull tops and higher stagnation pressures on the windward sides increase the net downward load. While this force is small compared with hydrodynamic lift, it tends to counteract dynamic squat slightly by pushing the boat back into the water.

Overall, the air-regime study shows that aerodynamic drag and vertical force increase with speed in a manner consistent with classical external-flow scaling. The drag and lift coefficients are nearly Reynolds-independent over the 20–60 km/h range because the above-water geometry is fixed and already operating at high Reynolds number. The resulting aerodynamic loads are one to two orders of magnitude smaller than the corresponding hydrodynamic loads, confirming that resistance and vertical support are dominated by water, not air, for straight-line running. However, the air loads are large enough to influence rigging design, mast and beam sizing, and dynamic stability in gusts, so including them in the overall performance assessment is justified.

CHAPTER 6. CONCLUSION

This project developed and analyzed a high-performance racing catamaran using an integrated Siemens NX workflow that combined parametric CAD, detailed material selection, linear static FEA, and steady RANS CFD. The goal was to understand how the Tornado-style platform behaves under representative—and in some cases intentionally conservative—operating conditions, and to assess whether the selected geometry and composite layups provide adequate strength, stiffness, and hydrodynamic performance.

From a structural standpoint, two limiting load cases were considered: a fresh gale with 40-knot winds and one-hull “flying” operation. In the fresh-gale scenario, aerodynamic loading on the sail was idealized as a flat plate in cross-flow and applied as a concentrated resultant at the mast–sail interface. After mesh refinement, the maximum von Mises stress in the mast reached approximately 608 MPa, converged at a 20 mm element size. This value lies within the reported 533–774 MPa yield range for the biaxial epoxy/HS carbon fiber laminate, but with essentially no safety margin once fabrication defects, impact damage, and stress concentrations are considered. A complementary hand calculation, modeling the mast as a uniform circular cantilever loaded at the tip, produced a still higher estimate (~1,090 MPa), reinforcing the conclusion that the rig is not structurally robust for sustained 40-knot conditions. These results are consistent with the Tornado’s role as a lightweight Olympic racing craft rather than an offshore heavy-weather design.

In the flying-a-hull case, the leeward hull and crossbeams carry the full weight of the vessel, crew, and rig while the windward hull is effectively suspended. Here, the maximum von

Mises stress in the crossbeam–hull region converged to roughly 250 MPa for a 50 mm mesh, well below the 383–552 MPa yield range of the quasi-isotropic epoxy/HS carbon fiber used for hulls and beams. A simple cantilever-beam hand model, based on measured component volumes and weights, predicted an outer-fiber bending stress of about 196 MPa—within 28 % of the FEA result and of the same order of magnitude. This level of agreement, along with the expected displacement pattern (largest deflection at the masthead and local stress concentrations near the beam–hull joints), supports the validity of the linear static model. Given a nominal factor of safety of 2 (~500 MPa allowable), the structure is close to the desired margin under highly conservative assumptions (fully fixed leeward hull, horizontal platform, ignored daggerboard support, and large deflections permitted below the waterline). In realistic sailing, shorter moment arms, dynamic relief from waves, and load sharing by the daggerboards would reduce peak stresses, suggesting that the design is adequate for aggressive—but not extreme—one-hull operation.

The CFD study complemented the structural analyses by quantifying hydrodynamic and aerodynamic loads over a 20–60 km/h speed range with a prescribed squat-versus-speed relationship. Using a refined TET4 mesh and an SST RANS model, the water-domain simulations captured the expected flow physics: stagnation at the bows, accelerated flow and suction along the shoulders, largely attached flow over the hulls and rudders, and a recovering wake downstream. Hydrodynamic drag increased from about 716 N at 20 km/h to roughly 3,030 N at 60 km/h, broadly following the V^2 scaling for turbulent external flow. At the same time, the drag and lift coefficients decreased substantially (C_D from ~136 to ~64 and C_L from ~127 to ~60) because the immersed volume and wetted area shrink with speed. This trend indicates that the

hull becomes effectively “cleaner” as it rides higher, with dynamic lift large enough to reduce draft and thereby mitigate further drag growth. Pressure extremes on the wetted surfaces scaled with dynamic pressure, confirming that the mesh and turbulence model captured the essential stagnation and suction behavior that drives both resistance and vertical support.

In the air domain, the same geometry and solver settings were used with only the above-water portion of the hulls, mast, and rigging exposed to flow. For the 60 km/h case, the solution converged cleanly, with velocity and pressure fields showing classical external-aerodynamic patterns around the bows, crossbeams, and mast. Aerodynamic drag increased from approximately 12 N at 20 km/h to about 99 N at 60 km/h, while C_D decreased only slightly ($0.95 \rightarrow 0.88$), reflecting weak Reynolds-number dependence for this fixed geometry. Vertical force remained small but non-negligible, with C_L rising from ~ 0.20 to ~ 0.26 and corresponding to a downward load on the hulls that slightly counteracts hydrodynamic lift. Overall, air loads were one to two orders of magnitude smaller than water loads at all speeds—air drag was only $\sim 3\%$ of total drag and the aerodynamic vertical force was $\sim 1\%$ of hydrodynamic support at 60 km/h—confirming that straight-line resistance and buoyancy are dominated by the water, while aerodynamic effects primarily influence rig sizing and dynamic stability.

Taken together, the FEA and CFD results paint a coherent picture of the Tornado-style catamaran as a structurally efficient but lightly margined racing platform. The composite layups chosen for hulls, beams, mast, and sails provide sufficient stiffness and strength for typical high-performance sailing and for conservative flying-a-hull scenarios, but approach or exceed allowable stresses under fresh-gale loading. Hydrodynamically, the hulls generate strong

dynamic lift that reduces draft and drag with increasing speed, while the above-water geometry contributes relatively modest aerodynamic drag and a small downward force that partially offsets squat. The close agreement between FEA and hand calculations, and the physically consistent trends in drag, lift, and pressure distributions from CFD, lend confidence to the modeling choices, even though several simplifying assumptions were made.

Future work could refine and extend this study in several directions. On the structural side, nonlinear laminate analysis, local ply-level failure criteria, and explicit modeling of joints and fittings would improve prediction of damage onset in extreme events. Including daggerboards, crew mass distribution, and realistic heel angles would give a more faithful representation of flying-hull load paths. On the fluid side, a coupled free-surface solver with sinkage and trim, unsteady RANS or DES for vortex shedding, and parametric variations in hull shape and beam spacing would enable genuine draft-angle optimization rather than the prescribed squat used here. Nonetheless, within the scope of ME4042, the present work demonstrates a robust, end-to-end workflow for racing catamaran design, linking parametric CAD, advanced composites, FEA, and CFD to produce an engineering-grounded assessment of performance and structural integrity.